

Solution



Step 4: Find the ratio of number of grains intersected to total length of the line i.e. n_L

In this problem length of each line is measured and is equal to 76.2mm

$$n_L = \frac{\text{Total number of intersections}}{\text{Total length of 5 lines}} = \frac{31}{5 \times 76.2} = 0.081 \text{ mm}^{-1}$$

Solution

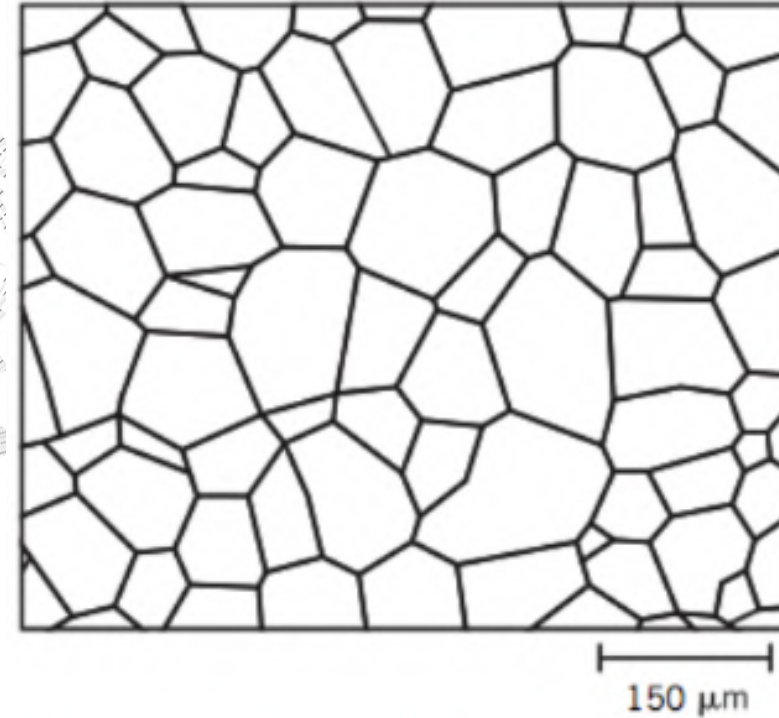


Step 5: Determine magnification of the image

$$M = \frac{\text{Measured value on the scale}}{\text{Printed value on the scale}}$$

$$M = \frac{25.5 \text{ mm}}{150 \mu\text{m}} = \frac{25.5 \times 1000 \mu\text{m}}{150 \mu\text{m}}$$

$$M = 170$$



Therefore magnification, M is equal to 170X

Solution



Step 6: Determine average grain size diameter

$$d = \frac{C}{n_L \times M}$$

Typical value of C is 1.5

$$d = \frac{C}{n_L \times M} = \frac{1.5}{0.81 \times 170} = 0.01089 \text{ mm} = 10.89 \text{ } \mu\text{m}$$

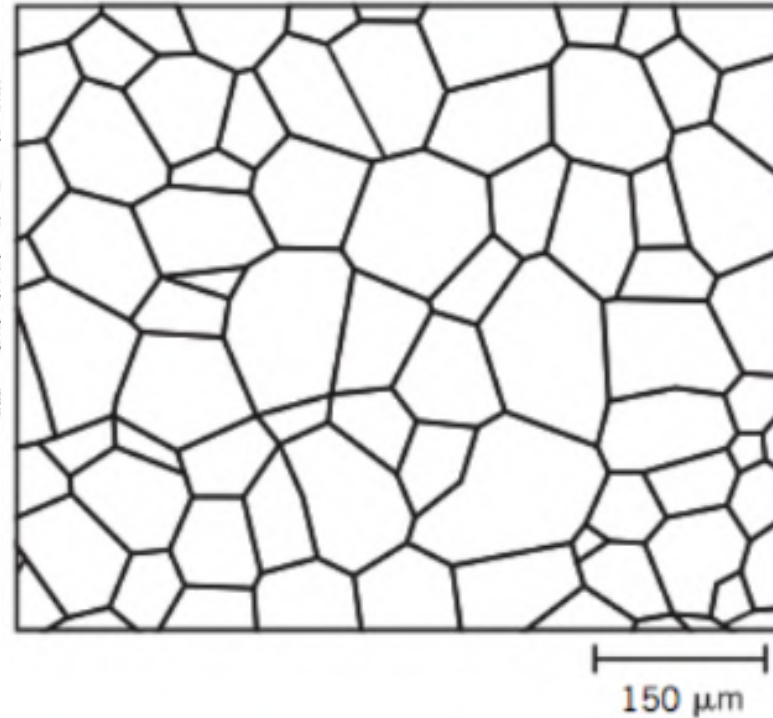
Solution



2. Calculate ASTM grain size number, n for this material

Step 1: Determine magnification of the image

For this diagram already magnification is determined. Magnification, M is equal to 170X



Solution

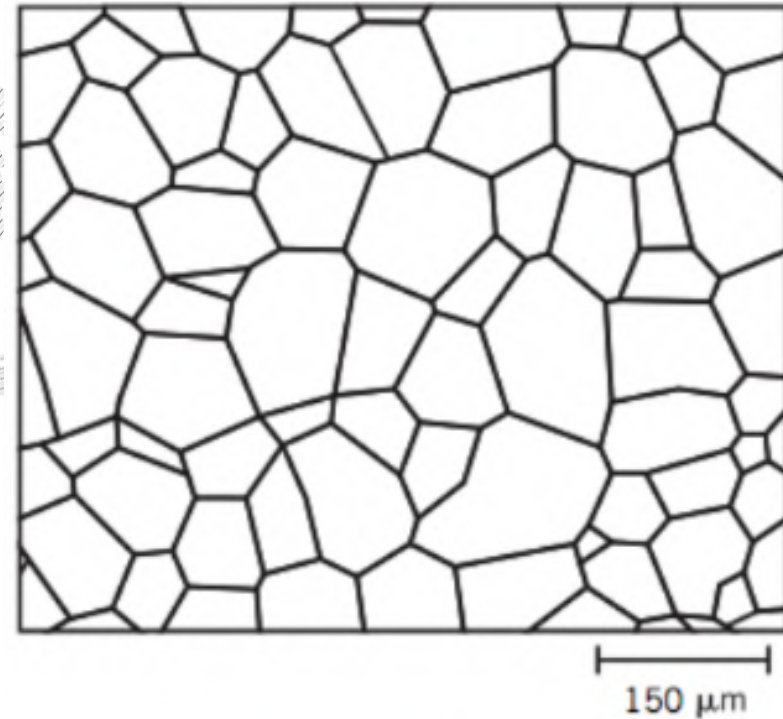


Step 2: Determine number of grains

**# of Grains =
Whole grains + (0.5 × partial grains)**

Whole grains are the grains which are within the boundary

Partial grains are out edges which are cropped.



Solution



$$\# \text{ of Grains} = 46 + 0.5 \times 35 = 63.5$$

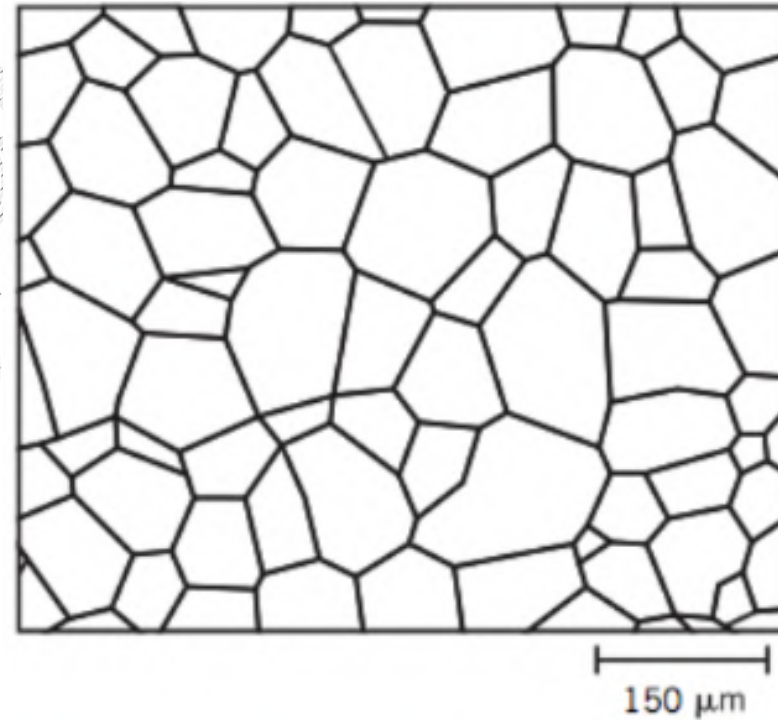
Step 3: Determine the true area

$$\text{True area} = \frac{W}{M} \times \frac{H}{M}$$

W is width of given micrograph.
In this case it is 115 mm

H is height of given micrograph.
In this case it is 95 mm

M is magnification. In this case it is 170X



Solution



$$\text{True area} = \frac{115}{170} \times \frac{95}{170} = 0.378 \text{ mm}^2$$

Step 4: Determine number of grain per sq. mm at 1X i.e. N'

$$N' = \frac{\text{Number of grains}}{\text{True area}} = \frac{63.5}{0.378} = 167.98 \frac{\text{grains}}{\text{mm}^2} \text{ at } 1X$$

Step 5: Determine number of grain per sq. inch at 100X i.e. N

$$N = \frac{N'}{15.5} = \frac{167.98}{15.5} = 10.84$$

Solution



Step 6: Determine ASTM grain size number, n

$$N = 2^{(n-1)}$$

$$\log(N) = (n-1)\log(2)$$

$$(n-1) = \frac{\log(N)}{\log(2)} = \frac{\log(10.84)}{\log(2)} = 3.44$$

$$n = 3.44 + 1 = 4.44 \approx 5$$

References



1. William D Calister Jr,” Material Science & Engineering, an Introduction”, John Wiley & Sons, New York. 8th edition
2. William F. Smith, Javad Hashemi, ‘Material Science & Engineering”, In SI units Fourth edition, McGraw-Hill.



Work Integrated Learning Programmes

Thank You



BITS Pilani

SEM and Applications

P Srinivasan
Core Engineering

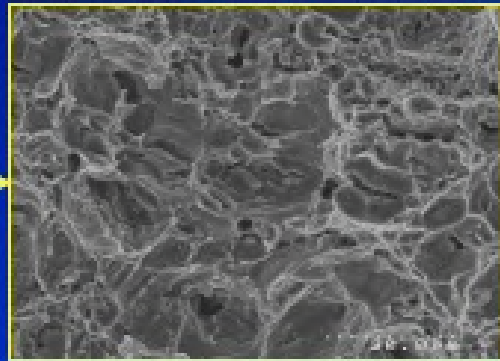


<Course Number, Course Title>
Lecture No.

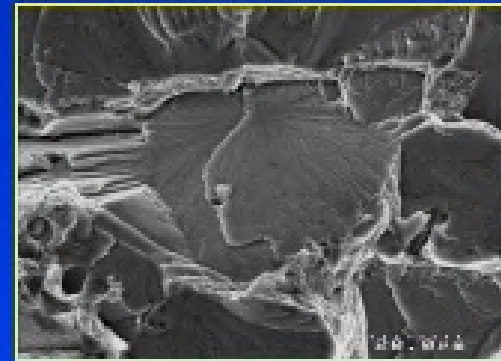
Types of Fracture



Failure modes



- **High energy** is absorbed by microvoid coalescence during ductile failure (high energy fracture mode)

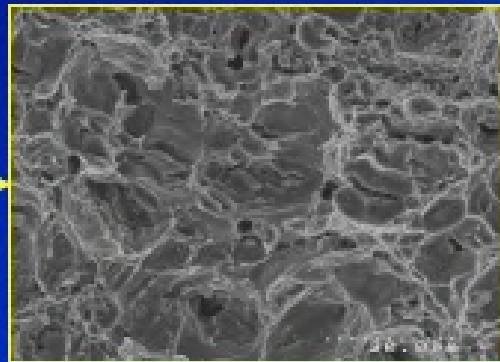


- **Low energy** is absorbed during transgranular cleavage fracture (low energy fracture mode)

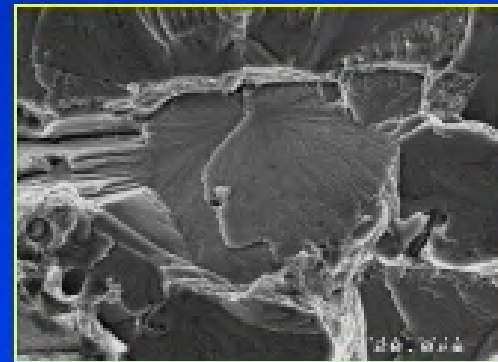
Types of Fracture



Failure modes

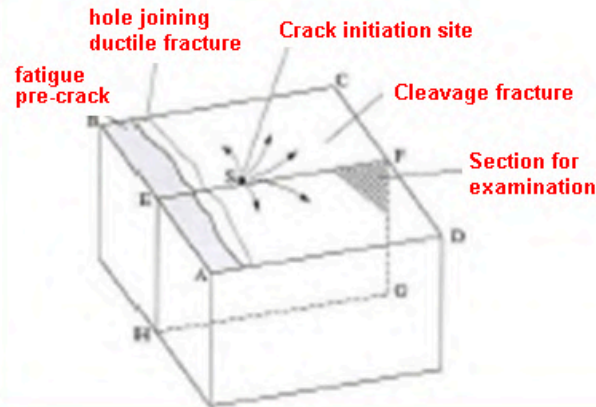
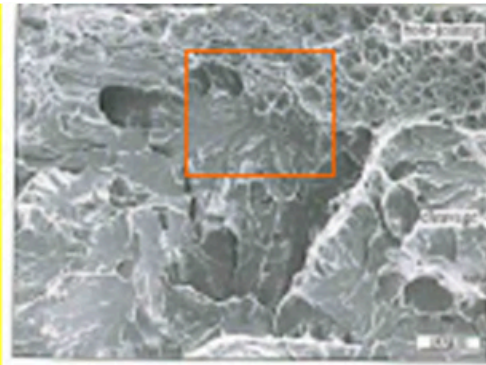


- **High energy** is absorbed by microvoid coalescence during ductile failure (high energy fracture mode)



- **Low energy** is absorbed during transgranular cleavage fracture (low energy fracture mode)

Types of Fracture



- Inclusions, porosity, second-phase particles or precipitates are preferential sites (**stress raiser**) for cleavage initiation.
- Fracture occurs along the **crystallographic planes**.
- The direction of the river pattern represents the **direction of the crack propagation**.

Types of Fracture



Microvoid formation, growth and coalescence

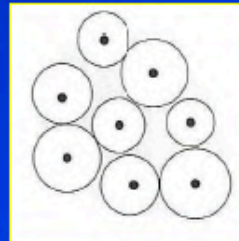
- **Microvoids** are easily formed at inclusions, intermetallic or second-phase particles and grain boundaries.
- **Growth** and **coalescence** of microvoids progress as the local applied load increases.



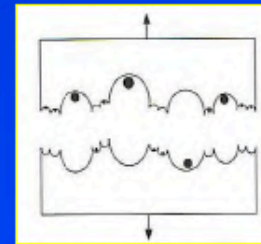
Ductile dimples centred on spherical particles



a) Random planar array of particles acting as void initiators.



b) Growth of voids to join each other as the applied stress increases.



c) Linkage or coalescence of these voids to form free fracture surface.

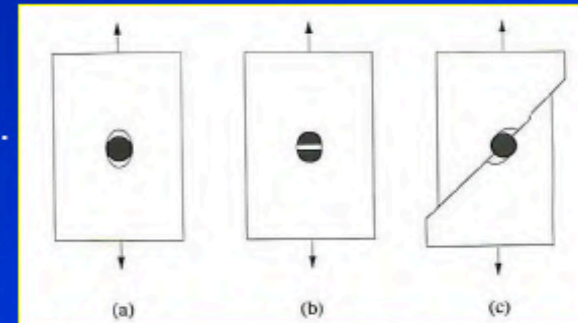
Types of Fracture



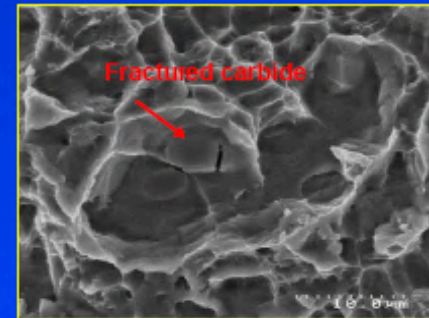
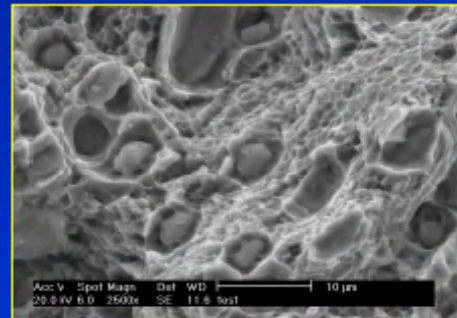
Formation of microvoids from second phase particles

Microvoids are formed by

- 1) Decohesion at particle-matrix interface.
- 2) Fracture of brittle particle
- 3) Decohesion of an interface associated with shear deformation or grain boundary sliding.



Mechanisms of microvoid formation

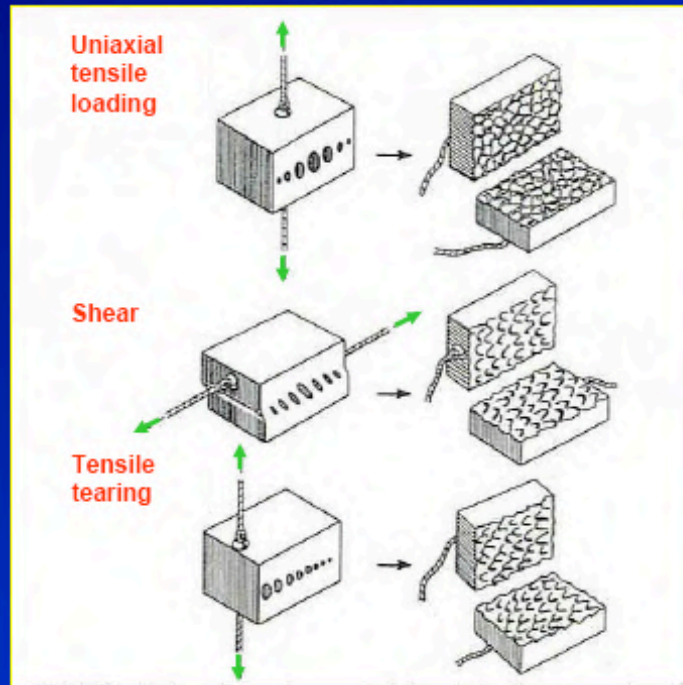


Types of Fracture



Microvoid shape

Microvoid shape is strongly influenced by the type of loading.



Uniaxial tensile loading

→ Equiaxed dimples.

Shear loading

→ Elongated and parabolic dimples pointing in the opposite directions on matching fracture surfaces.

Tensile tearing

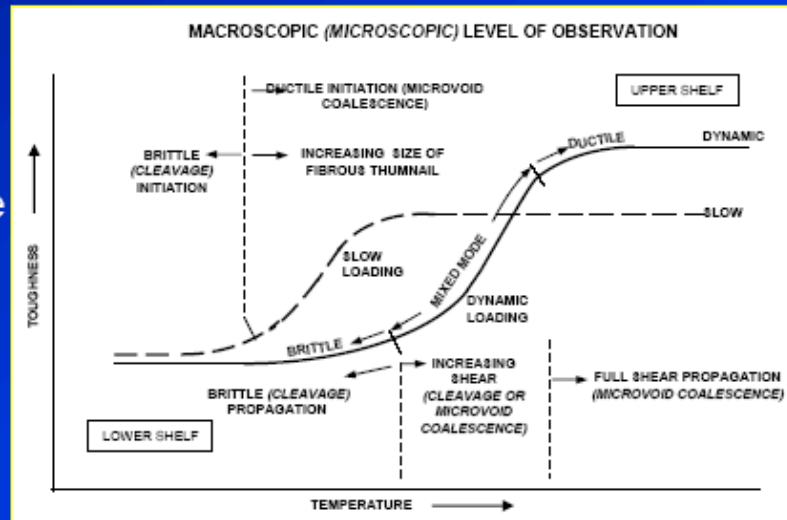
→ Elongated dimples pointing in the same direction on matching fracture surface.

Types of Fracture

Ductile to brittle transition behaviour

BCC structure metals experience **ductile-to-brittle transition behaviour** when subjected to decreasing temperature, resulting from a strong yield stress dependent on temperature.

- **BCC** metals possess **limited slip systems** available at low temperature, **minimising the plastic deformation** during the fracture process.
- **Increasing temperature** allows more slip systems to operate, yielding **general plastic deformation** to occur prior to failure.



Low temperature

High temperature

Brittle cleavage fracture

Ductile fracture

Types of Fracture

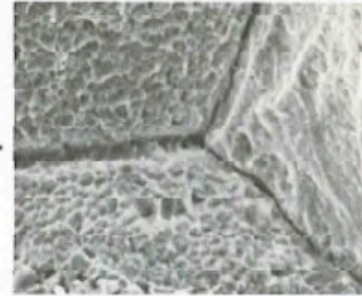


Intergranular fracture

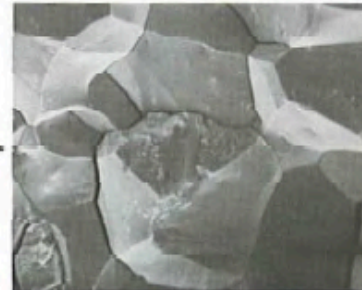
- **Intergranular failure** is a moderate to low energy brittle fracture mode resulting from **grain boundary separation** or segregation of embrittling particles or precipitates.

- **Embrittling grain boundary particles** are weakly bonded with the matrix, → high free energy and unstable, which leads to **preferential crack propagation path**.

Intergranular fracture with microvoid coalescence



Intergranular fracture without microvoid coalescence



Intergranular fracture with and without microvoid coalescence.

Types of Fracture



Factors affecting modes of fracture

Metallurgical aspect

Temperature

**State of stresses
(notch effect)**

Strain rate

Loading condition

Brittle fracture

Large grained materials
with GB particles.

Low temperature

Triaxial state of
stresses (notch effect)

High strain rate

Ductile fracture

Fine grained material
without GB particles.

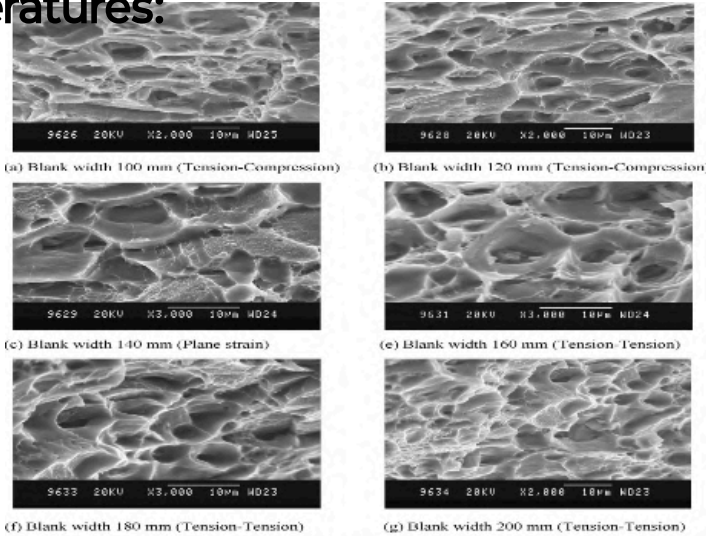
High temperature

Absence of the notch

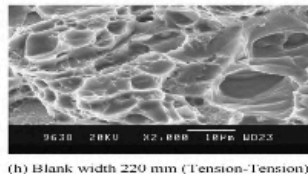
Low strain rate

Hydrostatic pressure
(suppress crack initiation)

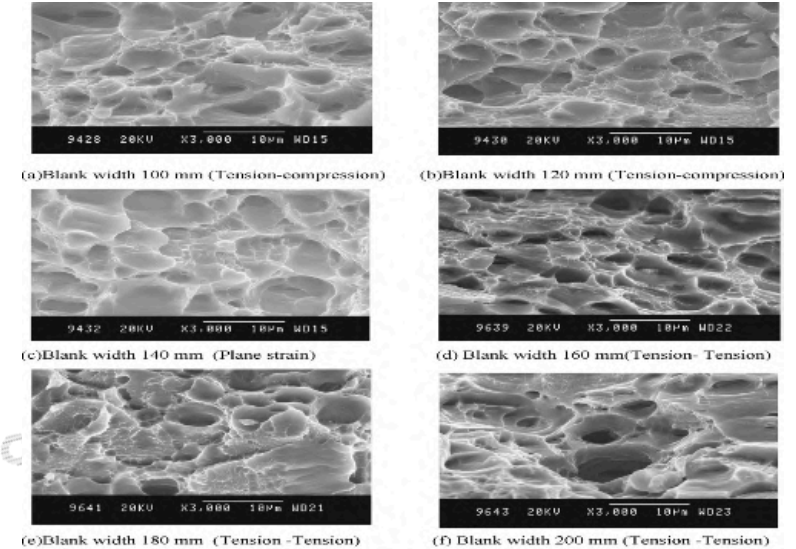
◦ SEM images taken for the fracture surfaces for Al 5052 sheets annealed at different temperatures:



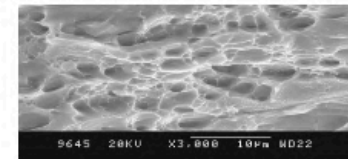
a. 200 °C.



(h) Blank width 220 mm (Tension-Tension)



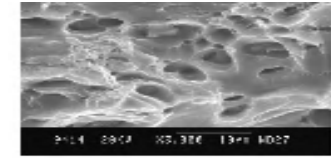
b. 250 °C.



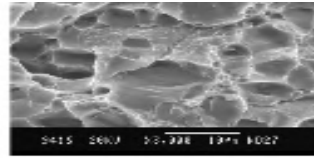
(g) Blank width 220 mm (Tension-Tension)

► The sheets annealed at 350 °C shows large number of voids in the SEM images taken at its fracture surface compared to rest of the other three annealing temperatures, viz. 200 , 250 and 300 °C.

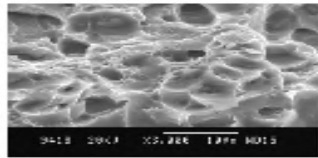
o SEM images taken for the fracture surfaces for Al 5052 sheets annealed at different temperatures:



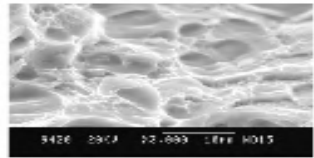
(a) Blank width 100 mm (Tension-compression)



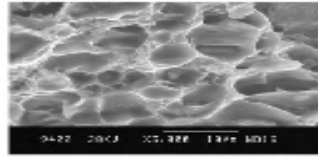
(b) Blank width 120 mm (Tension-compression)



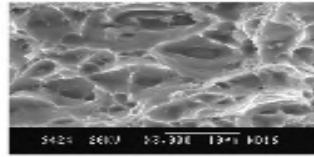
(c) Blank width 140 mm (Plane strain)



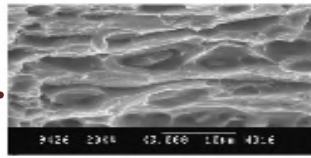
(d) Blank width 160 mm (Tension-Tension)



(e) Blank width 180 mm (Tension-Tension)

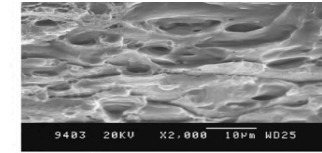


(f) Blank width 200 mm (Tension-Tension)

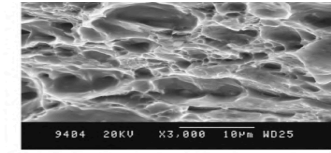


(g) Blank width 220 mm (Tension-Tension)

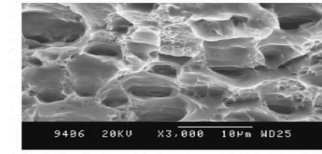
c. 300 °C.



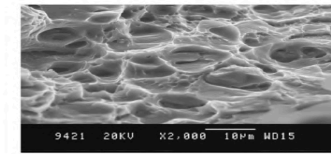
(a) Blank width 100 mm (Tension-compression)



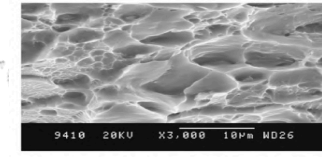
(b) Blank width 120 mm (Tension-compression)



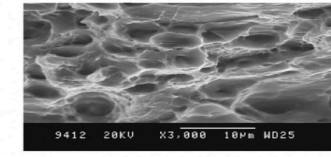
(c) Blank width 140 mm (Plane strain)



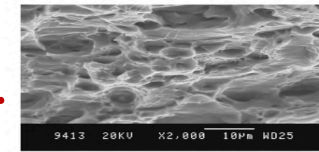
(d) Blank width 160 mm (Tension-Tension)



(e) Blank width 180 mm (Tension-Tension)



(f) Blank width 200 mm (Tension-Tension)

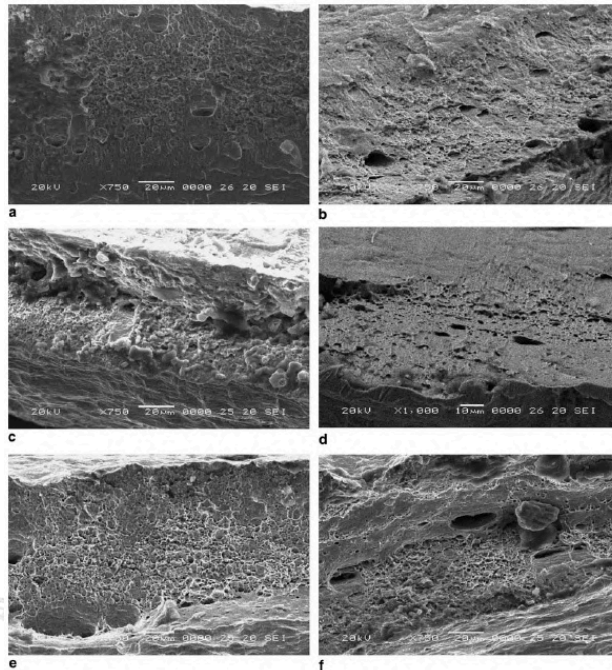


(g) Blank width 220 mm (Tension-Tension)

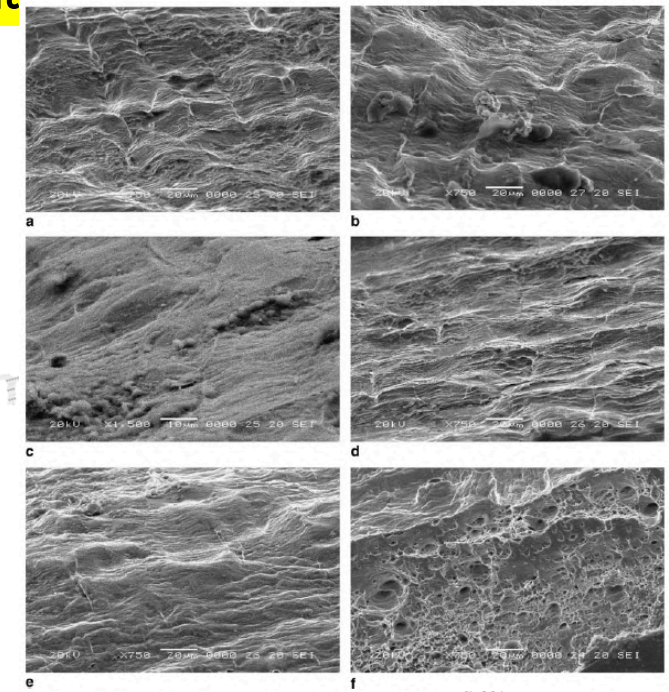
d. 350 °C.

► This large number of voids is because of the accommodation of high plastic strain by the presence of fully or partially recrystallized grains in the annealed microstructure of Al 5052.

oSEM images of various blanks of IF steel of thickness 0.6 mm and 1.6 mm (a–c) Fracture surface for tension–compression conditions at 750x magnification. (d) Fracture surface for plane strain condition at 750x magnification. (e and f) Fracture surface for tension–tension conditions at 750x magnification



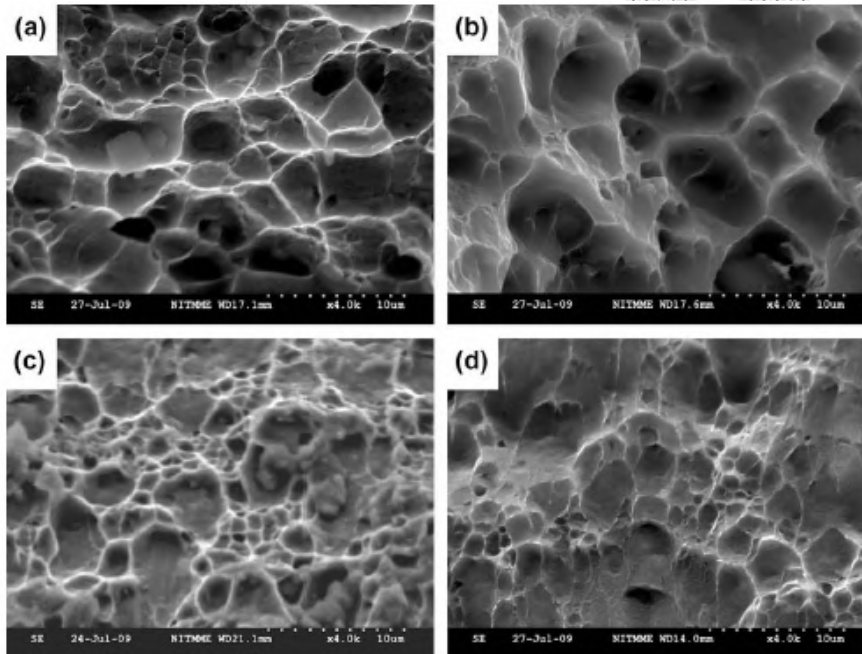
1. Thickness 0.6 mm



2. Thickness 1.6 mm

Studies on void coalescence analysis of nanocrystalline cryorolled commercially pure aluminium formed under different stress conditions:

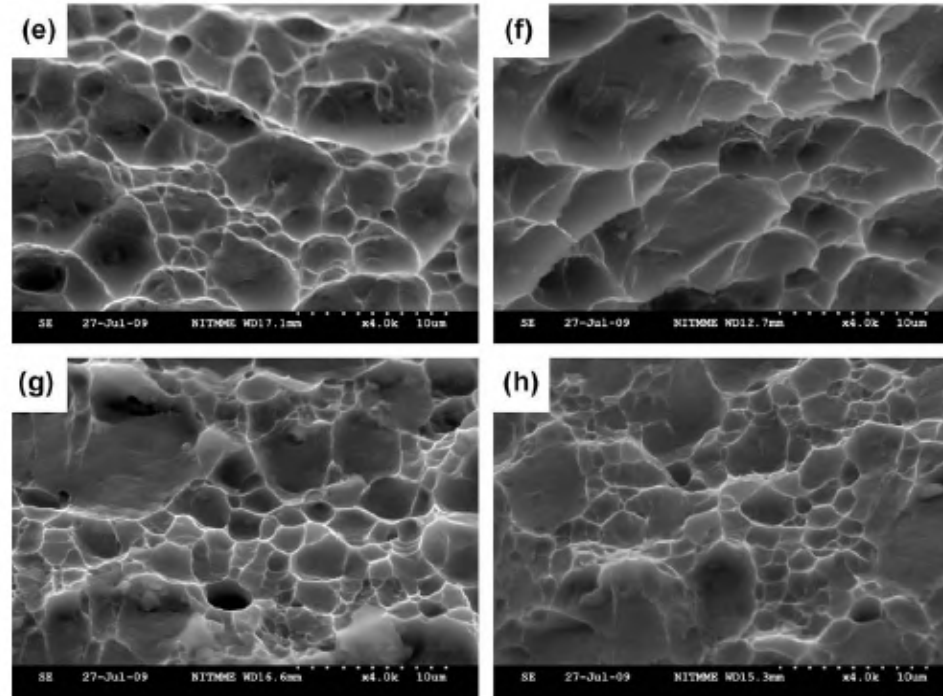
► Fractographs of cryorolled commercially pure aluminium of 0.25 mm thickness:



(a) notch radius of 1.0 mm;
(b) notch radius of 1.5 mm;

(c) notch radius of 2.0 mm;
(d) notch radius of 2.5 mm .

Continue....



(e) notch radius of 3.0 mm; (f) notch radius of 3.5 mm;

(g) notch radius of 4.0 mm; (h) notch radius of 5.0 mm.

o TEM micrograph in cryorolled condition:

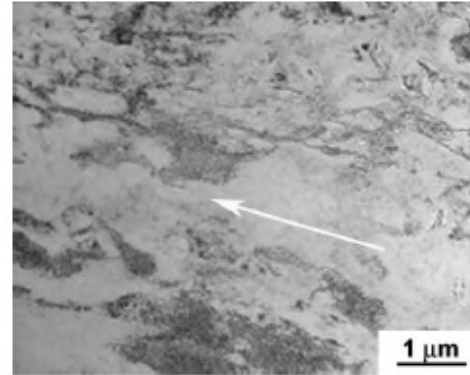
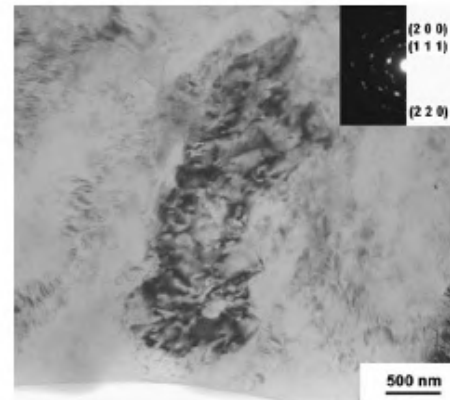


Fig. 2. TEM micrograph in cryorolled condition.

o TEM micrograph in cryorolled condition exhibiting severe dislocation cell structure; inset exhibits ring pattern with streaks:



► Fractography of Al-19000 annealed at 300°C at 3000x

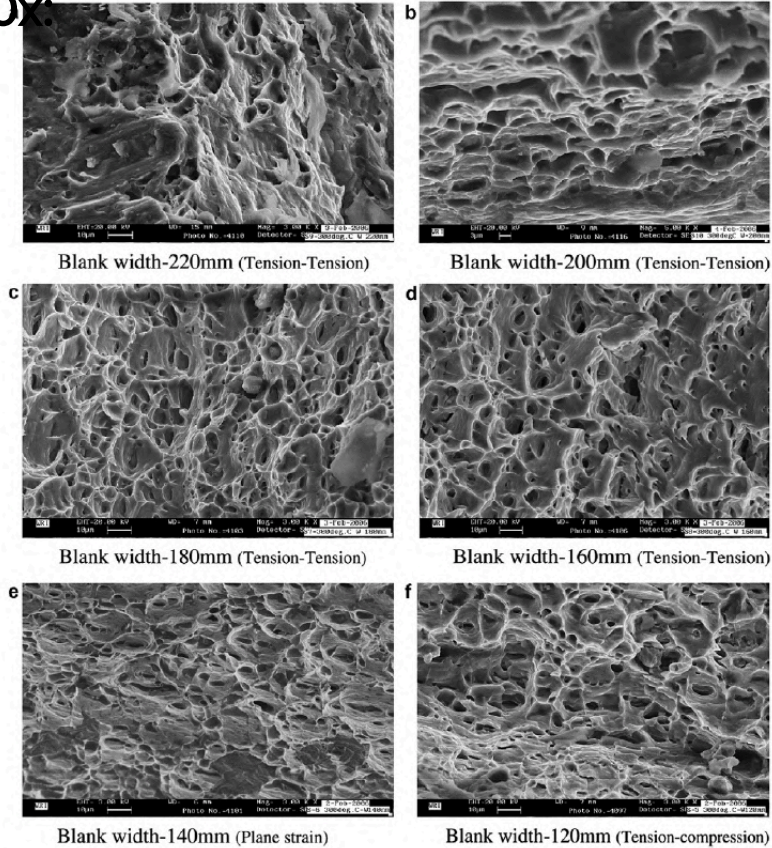
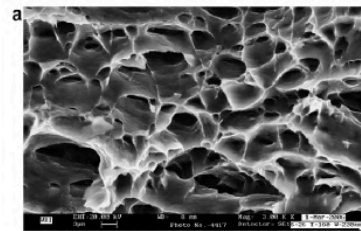
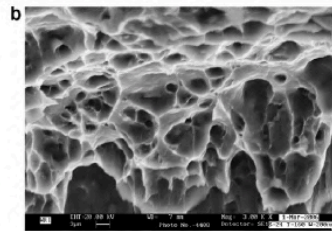


Fig: (a) Blank width 220 m (Tension–Tension), (b) blank width 200 m (Tension–Tension), (c) blank width 180 m (Tension–Tension), (d) blank width 160 m (Tension–Tension), (e) blank width 140 m (Plane strain) and (f) blank width 120 m (Tension–Compression).

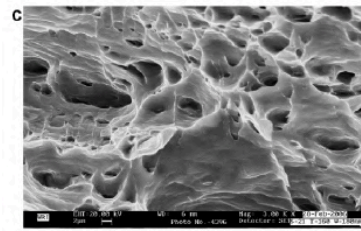
► Fractography of Al-19000 annealed at 160°C at 3000x:



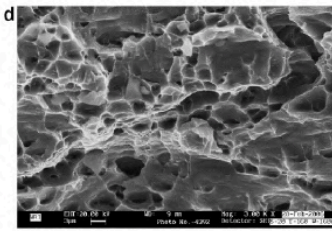
Blank width 220mm (Tension-Tension)



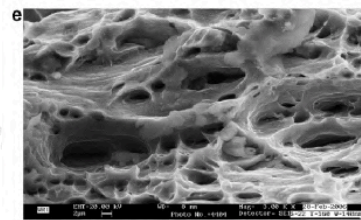
Blank width 200mm (Tension-Tension)



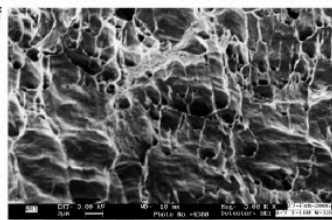
Blank width 180mm (Tension-Tension)



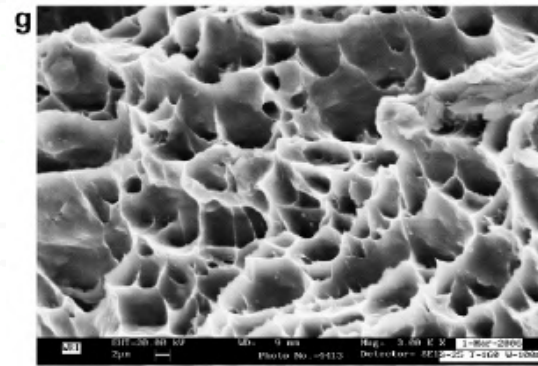
Blank width 160mm (Tension-Tension)



Blank width 140mm (Plane strain)



Blank width 120mm (Tension- compression)



Blank width 100mm (Tension- compression)

Fig: (a) Blank width 220 m (Tension–Tension), (b) blank width 200 m (Tension–Tension), (c) blank width 180 m (Tension–Tension), (d) blank width 160 m (Tension–Tension), (e) blank width 140 m (Plane strain) and (f) blank width 120 m (Tension–Compression), and (g) blank width 100 m (Tension–Compression).



BITS Pilani
Pilani Campus

Material Technology & Testing (DE ZG631)

Prof . P. Srinivasan





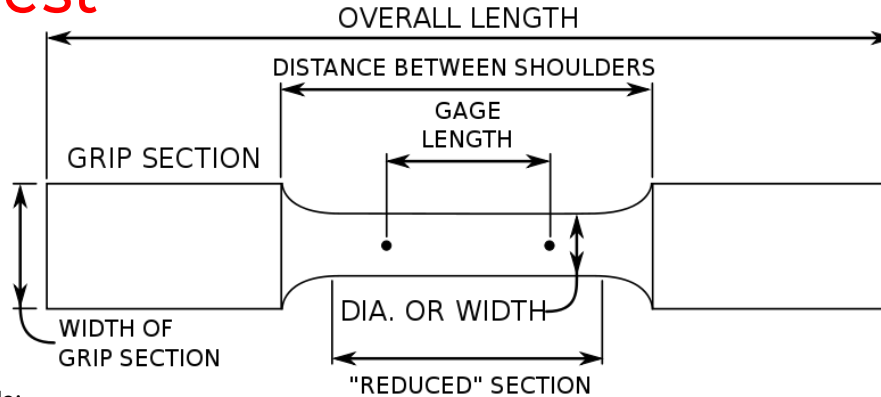
Tensile Test



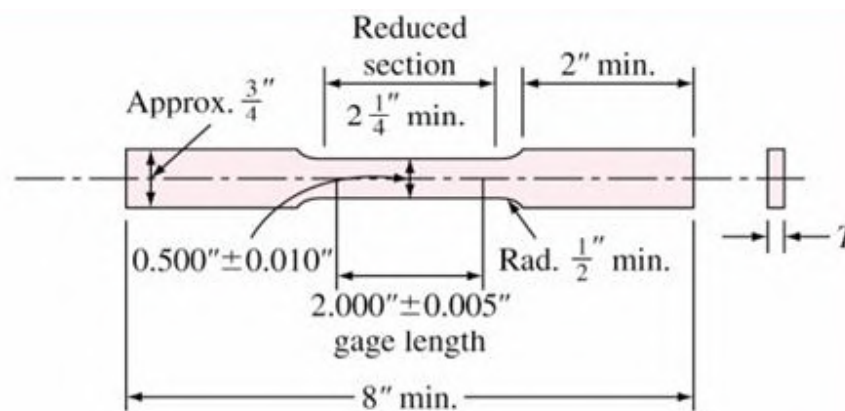
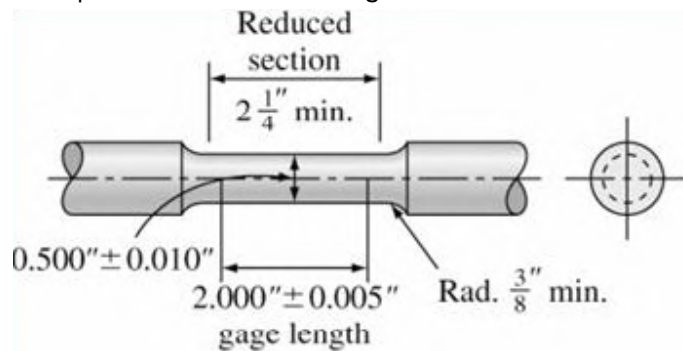
The Tension Test

- ❑ It is most important test as far as mechanical properties of a material is concern
- ❑ The test is conducted till specimen fractures or ruptures
- ❑ Therefore this test comprises elastic as well plastic deformation of the material.

The Tension Test



https://commons.wikimedia.org/wiki/File:Tensile_specimen_nomenclature.svg

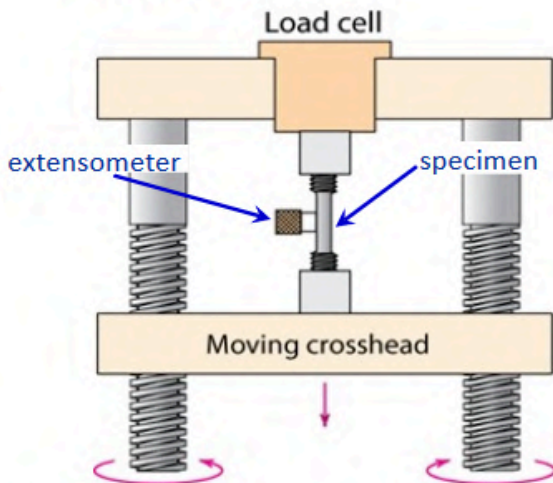


<http://pubs.sciepub.com/ajcea/2/1/6/>

The Tension Test

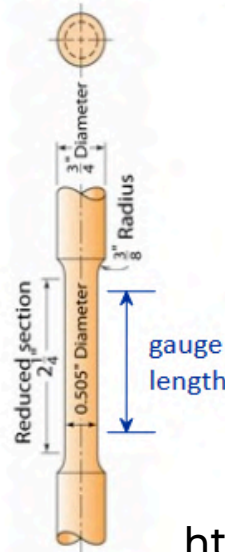
Stress-Strain: Testing Uses Standardized methods developed by ASTM for Tensile Tests it is ASTM E8

• Typical tensile test machine



Adapted from Fig. 6.3, Callister 7e. (Fig. 6.3 is taken from H.W. Hayden, W.G. Moffatt, and J. Wulff, *The Structure and Properties of Materials*, Vol. III, *Mechanical Behavior*, p. 2, John Wiley and Sons, New York, 1965.)

• Typical tensile specimen (ASTM A-bar)



- In this test a material is stretched at a specific rate.
- the force required to cause an extension is measured with a load cell.
- the extension is measured by an extensometer.
- stress and strain are calculated from force and deformation.

<http://slideplayer.com/slide/3815756/>

Tensile test

- Strength of materials can be tested by pulling the metal to failure.

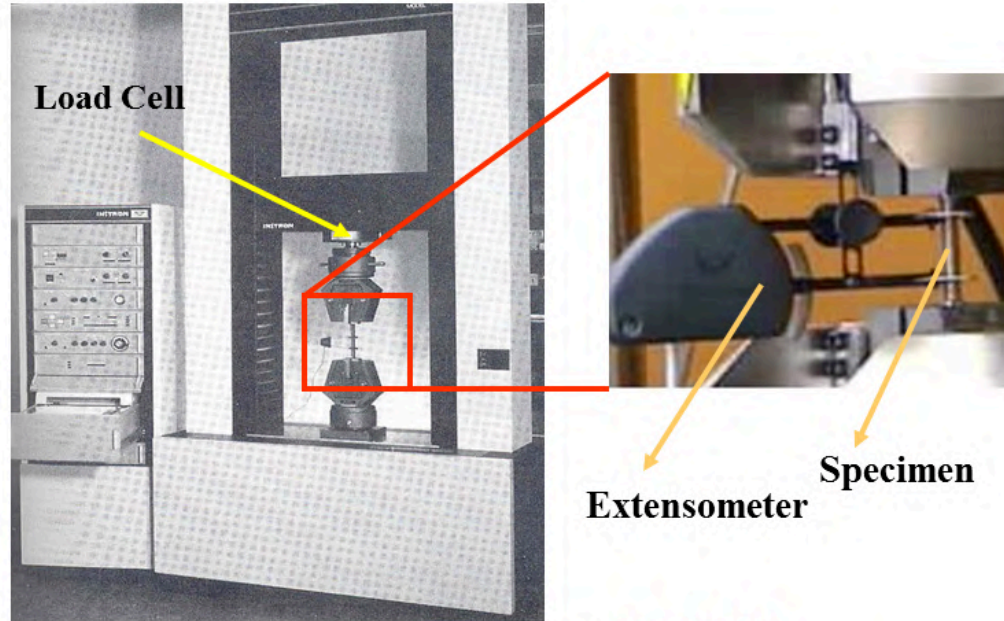
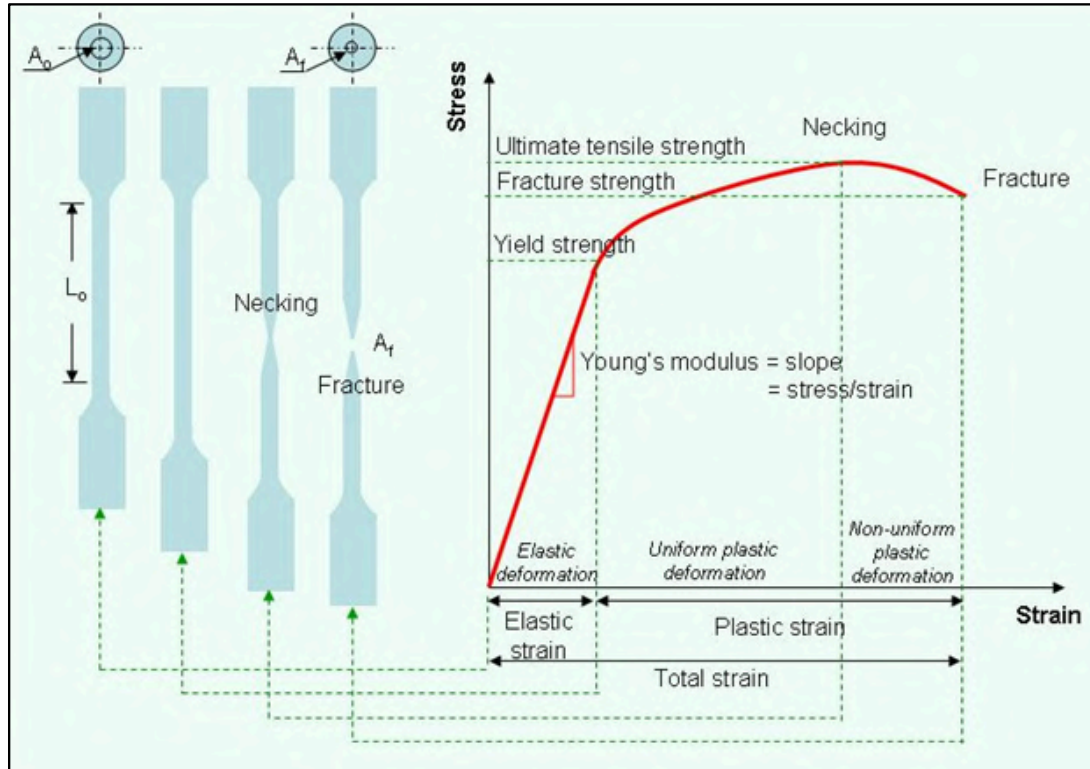


Figure 5.18

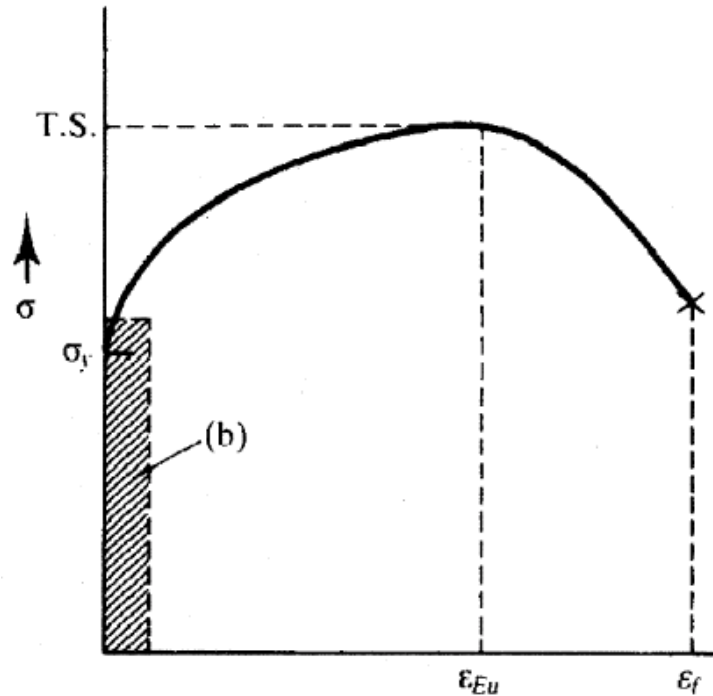
Force data is obtained from **Load cell**
Strain data is obtained from **Extensometer**.

The Tension Test

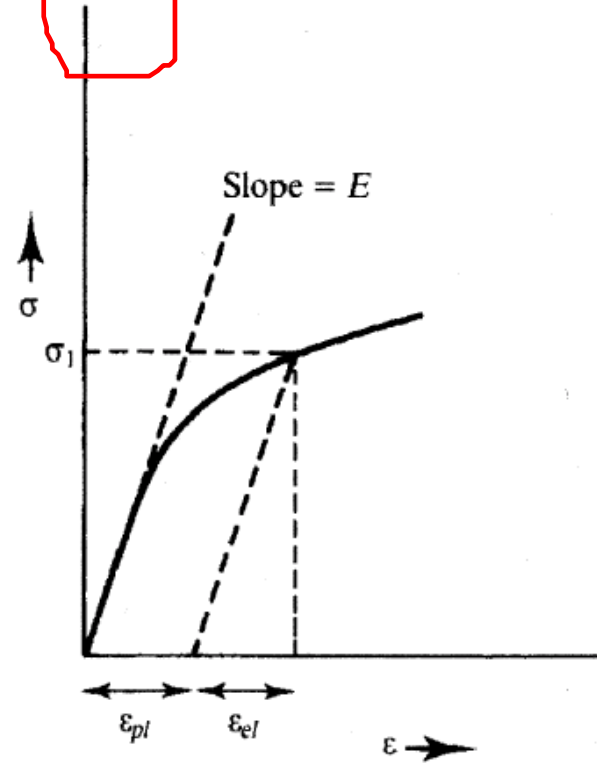


<http://pubs.sciepub.com/ajcea/2/1/6/>

The Tension Test



(a)



(b)

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- ❑ The subscript T is used to designate true stress (or strain) and the subscript E to denote engineering stress (or strain).

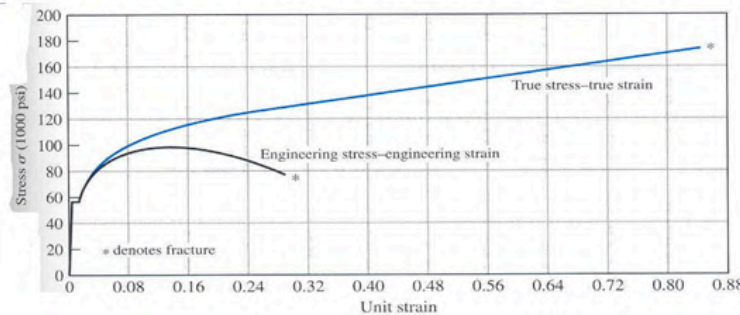
$$\sigma_T = \frac{F}{A_i}$$
$$\sigma_E = \frac{F}{A_o}$$

Engineering Stress (σ_E) & True Stress (σ_T)



True Stress – True Strain

- True stress and true strain are based upon **instantaneous** cross-sectional area and length.



- True Stress = $\sigma_t = \frac{F}{A_i \text{ (instantaneous area)}}$

- True Strain = $\epsilon_t = \int_{\ell_0}^{\ell_i} \frac{d\ell}{\ell} = Ln \frac{\ell_i}{\ell_0} = Ln \frac{A_0}{A_i}$

- True stress is **always greater** than engineering stress.

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Tensile Test (Cont)

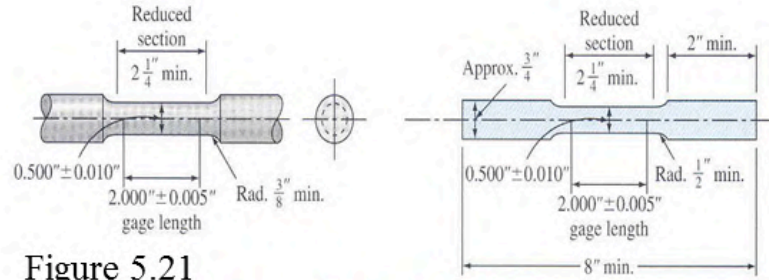
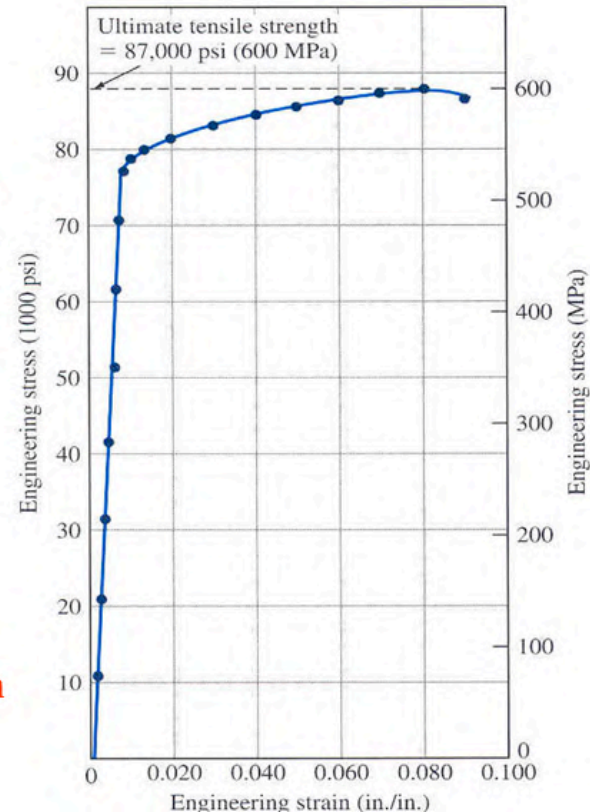


Figure 5.21

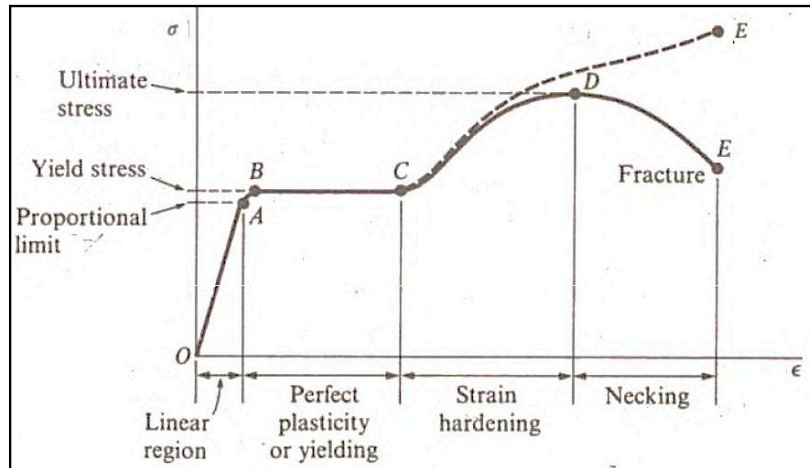
**Commonly used
Test specimen**

**Typical Stress-strain
curve**

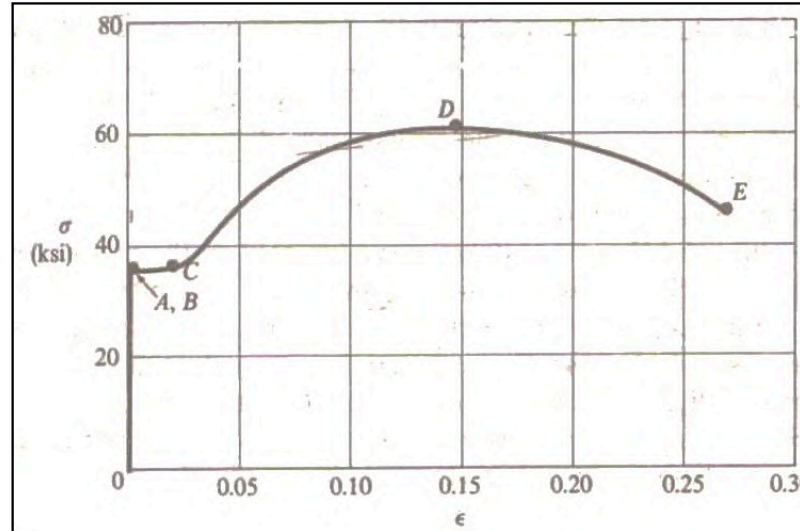


Stress- Strain diagram

Stress-strain diagram for a structural steel in tension



Not to
scale

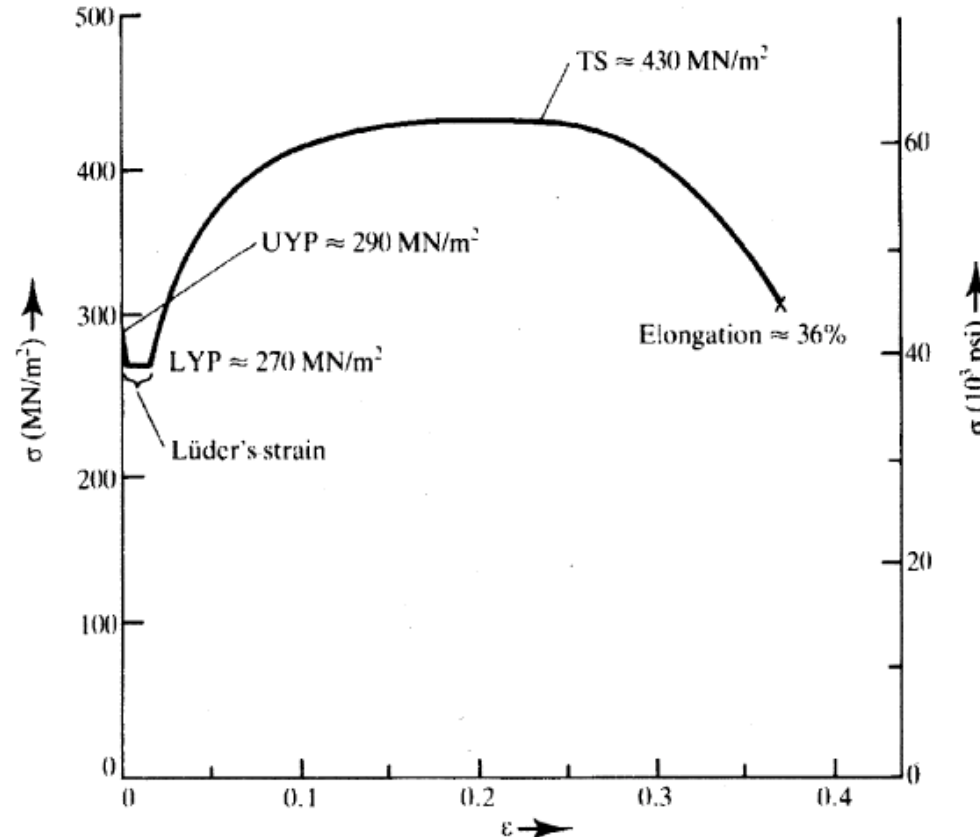


Draw to scale

Stress- Strain diagram

Engineering stress-strain diagram for a mild steel (1018 hot-rolled).

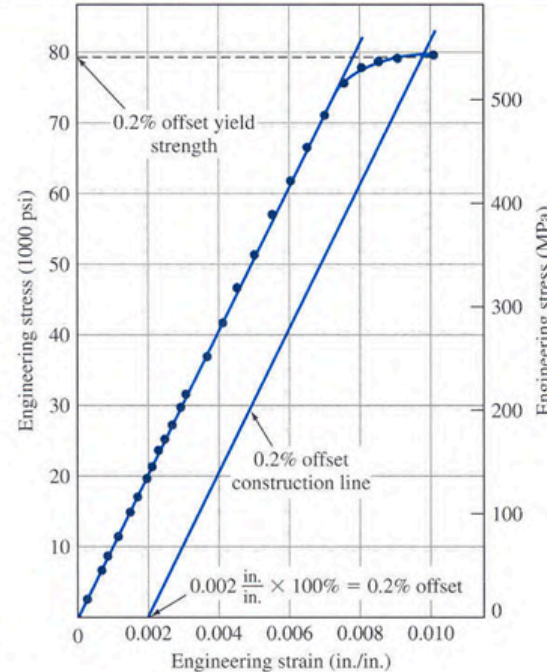
- ❑ The room temperature engineering stress strain curve of a mild steel is characterized by a yield point.





Yield Strength

- Yield strength is strength at which metal or alloy show significant amount of **plastic deformation**.
- 0.2% offset yield strength** is that strength at which 0.2% plastic deformation takes place.
- Construction line, starting at 0.2% strain and parallel to elastic region is drawn to find 0.2% offset yield strength.





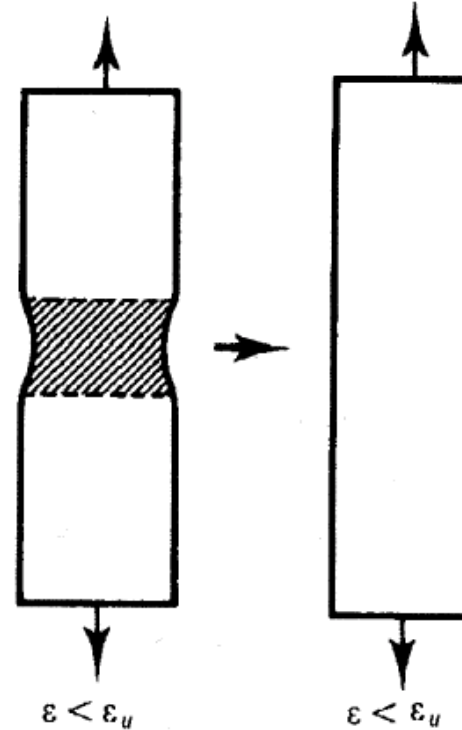




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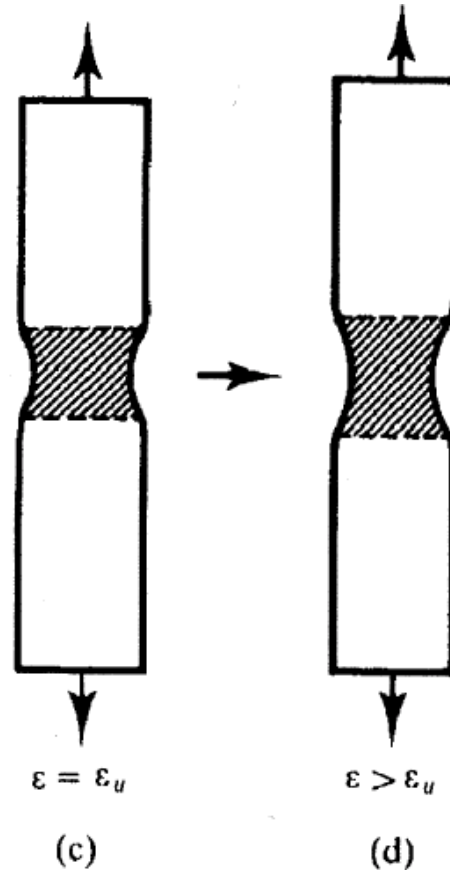
- ❑ For strains less than ϵ_u , the increase in flow stress accompanying the greater strains is large enough to lead to removal of the incipient instability.
- ❑ This process occurs regularly and repeatedly during tensile loading.



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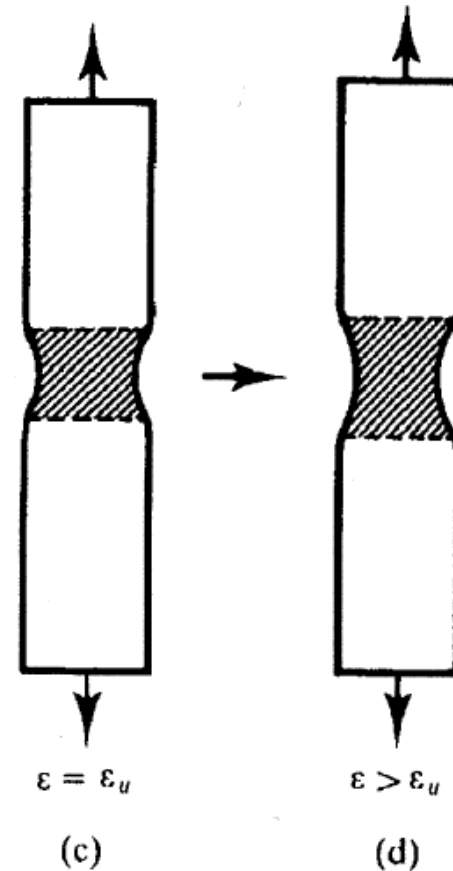
- ❑ The rate of work hardening decreases as deformation continues.
- ❑ Thus, it becomes progressively more difficult to work harden an incipient instability sufficiently to remove it.



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- ❑ At the tensile or ultimate point, the work-hardening capacity has been diminished enough that an instability once formed continues to develop.
- ❑ The criterion for necking can be expressed quantitatively, when the force reaches a maximum, i.e., $dF = 0$.
- ❑ Using the definition of true stress



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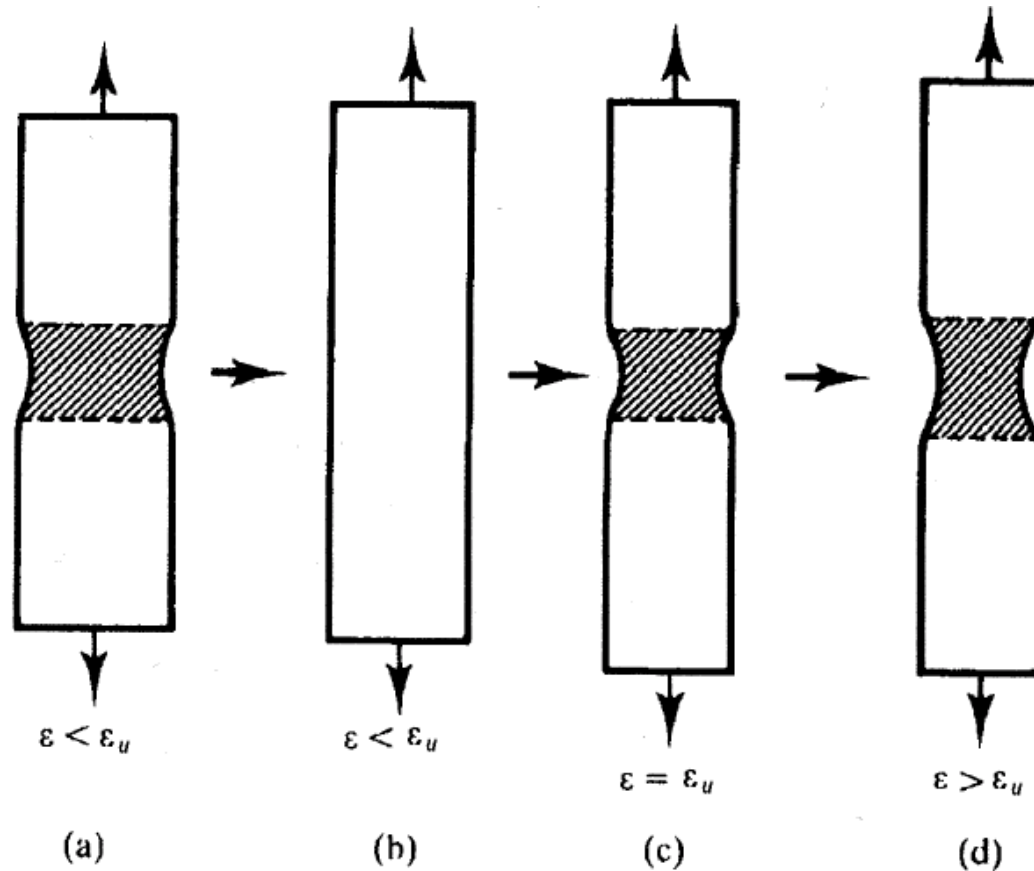


$$dF = 0 = \sigma_T dA_i + A_i d\sigma_T$$

$$\frac{d\sigma_T}{\sigma_T} = -\frac{dA_i}{A_i}$$

- i.e. the onset of necking is characterized by the fractional increase in f bw stress ($d\sigma_T/\sigma_T$) being exactly balanced by the fractional decrease in load-bearing area (dA_i/A_i).
- Prior to necking, $|d\sigma_T/\sigma_T| > |dA_i/A_i|$ and incipient instabilities are removed.
- Following necking, the instability continues to develop.

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Parameter	Fundamental definition	Prior to necking	Following necking
Engineering stress (σ_E)	$\sigma_E = \frac{F}{A_0}$	$\sigma_E = \frac{F}{A_0}$	$\sigma_E = \frac{F}{A_0}$
True stress (σ_T)	$\sigma_T = \frac{F}{A_i}$	$\sigma_T = \frac{F}{A_i}$ $= \sigma_E(1 + \epsilon_E)$	$\sigma_T = \frac{F}{A_{\text{neck}}}$
Engineering strain (ϵ_E)	$\epsilon_E = \frac{\delta l}{l_0}$	$\epsilon_E = \frac{\delta l}{l_0}$	$\epsilon_E = \frac{\delta l}{l_0}$
True strain (ϵ_T)	$\epsilon_T = \ln \frac{A_0}{A_{\text{min}}}$	$\epsilon_T = \ln \frac{l_i}{l_0}$ $= \ln \frac{A_0}{A_i}$ $= \ln (1 + \epsilon_E)$	$\epsilon_T = \ln \frac{A_0}{A_{\text{neck}}}$



Percent Elongation

- Percent elongation is a measure of **ductility** of a material.
- It is the elongation of the metal before fracture expressed as **percentage of original length**.

$$\% \text{ Elongation} = \frac{\text{Final length} - \text{initial Length}}{\text{Initial Length}}$$

- Measured using a caliper fitting the fractured metal together.
- **Example:-** Percent elongation of pure aluminum is 35%
For 7076-T6 aluminum alloy it is 11%



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References

1. William F. Smith, Javad Hashemi, 'Material Science & Engineering", In SI units Fourth edition, McGrawHill.
2. Mechanical Behavior of Materials by Thomas H. Courtney



Tensile Test of Polymers

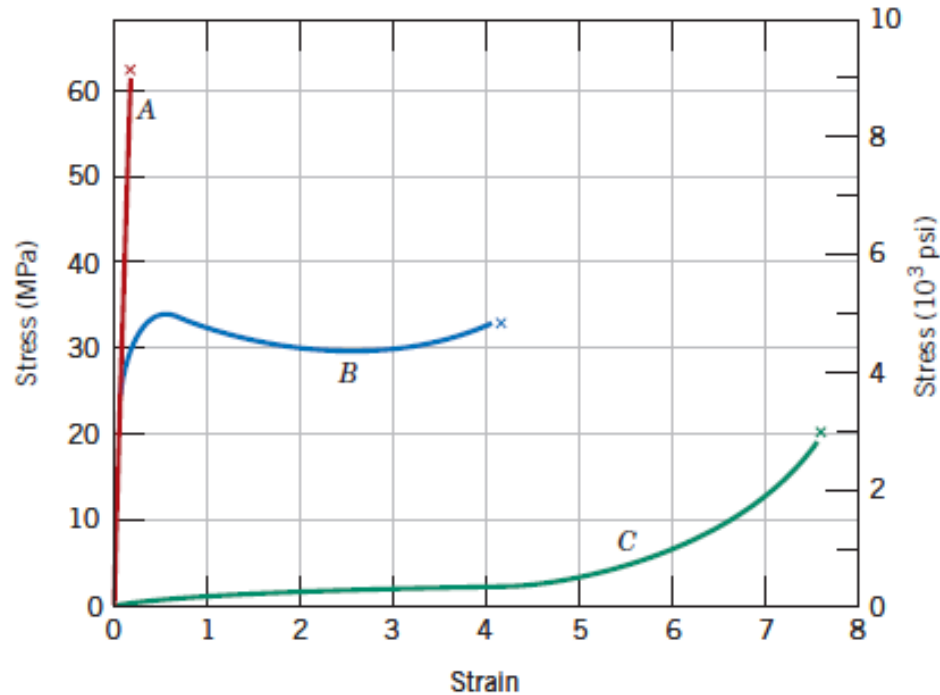
Reference

William F. Smith, Javad Hashemi, 'Material Science & Engineering', In SI units Fourth edition, McGrawHill.

➤ Chapter 10: Polymeric Materials

❑ Topic 10.10 Deformation & Strengthening of Plastic Materials

Mechanical Properties



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Mechanical Properties

Modulus for polymer ----- 7 MPa - 4 GPa

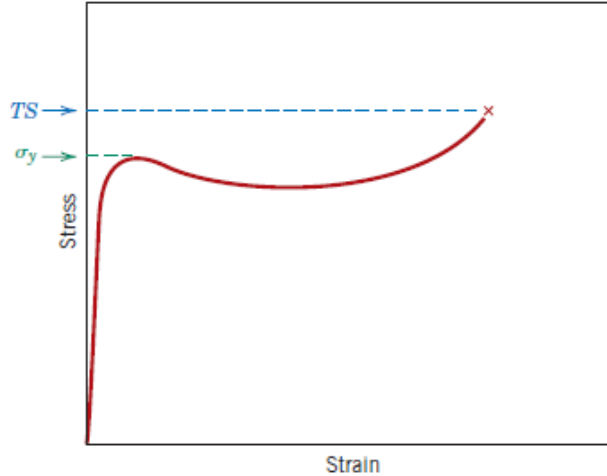
Modulus values for metals ----- 48GPa - 410 GPa

Maximum tensile strengths for polymers @100 MPa

For metal alloys up to 4100 MPa

metals rarely elongate plastically to more than 100%,
some highly elastic polymers may experience elongations to greater than 1000%.

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- For plastic polymers, the yield point is taken as a maximum on the curve, which occurs just beyond the termination of the linear-elastic region. The stress at this maximum is the yield strength.
- Tensile strength (TS) corresponds to the stress at which fracture occurs.
- TS may be greater than or less than yield strength.

Deformation of Thermoplastics

- Below T_g → Elastic deformation. Above T_g → Plastic deformation.

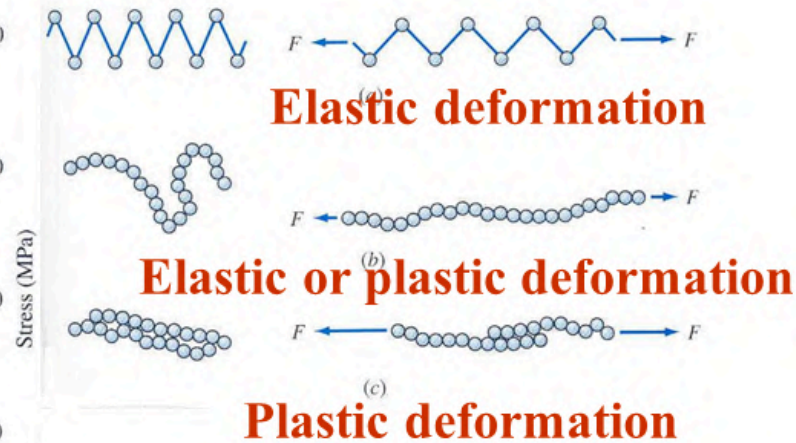
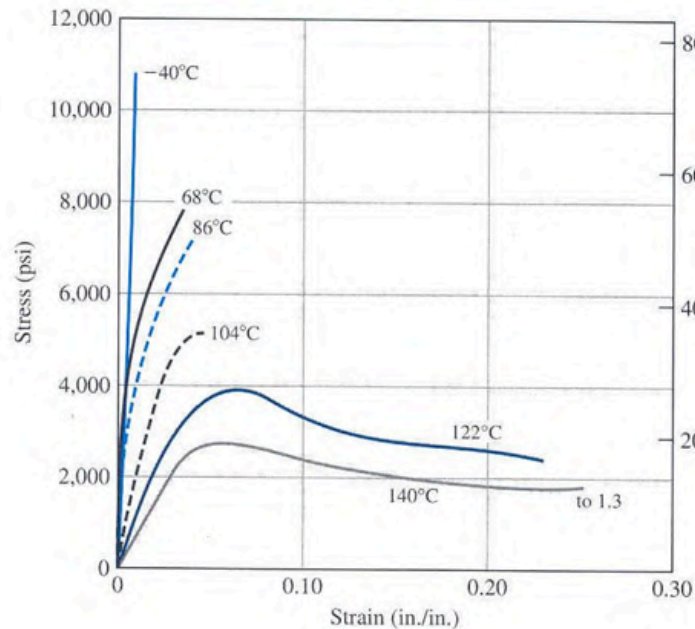


Figure 7.46



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Exercise Problems







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A 12.5 mm-diameter bar is subjected to a load of 2500 kg. Calculate the engineering stress on the bar in Megapascals (MPa).

■ Solution

The load on the bar has a mass of 2500 kg. In SI units the force on the bar is equal to the mass of the load times the acceleration of gravity (9.81 m/s^2), or

$$F = ma = (2500 \text{ kg})(9.81 \text{ m/s}^2) = 24,500 \text{ N}$$

The diameter d of the bar = 12.5 mm = 0.0125 m. Thus, the engineering stress on the bar is

$$\begin{aligned}\sigma &= \frac{F}{A_0} = \frac{F}{(\pi/4)(d^2)} = \frac{24,500 \text{ N}}{(\pi/4)(0.0125 \text{ m})^2} \\ &= (2.00 \times 10^8 \text{ Pa}) \left(\frac{1 \text{ MPa}}{10^6 \text{ Pa}} \right) = 200 \text{ MPa} \blacktriangleleft\end{aligned}$$



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Ex.

If true stress-strain curve from the onset of plastic deformation to the point at which necking begins is approximated as

$$\sigma_T = K \epsilon_T^n$$

where K and n are material constants.

Determine K and n for the following true stresses produce the corresponding true plastic strains for steel

True Stress (MPa)	True Strain
350	0.10
415	0.20

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Sol.

$$\sigma_T = K \varepsilon_T^n \quad \Rightarrow \quad \log \sigma_T = \log K + n \log \varepsilon_T$$

$$\log(350) = \log K + n \log(0.10) \text{ --- (1)} \quad \log(415) = \log K + n \log(0.20) \text{ --- (2)}$$

By solving eq (1) & eq (2) for n

$$n = \frac{\log(350) - \log(415)}{\log(0.10) - \log(0.20)} = 0.246 \quad \text{and}$$

$$\log K = \log(415) - 0.246 \log(0.20) = 2.79 \quad \Rightarrow \quad K = 616.35 \text{ MPa}$$















Hardness

Brinell test

- ❑ It is a common hardness test.
- ❑ the indenter is a 10-mm diameter hardened steel ball or tungsten carbide ball.
- ❑ The applied mass is usually 3000 kg (for soft material 500kg)
- ❑ Load is applied for 30 seconds

Brinell hardness number (BHN) is expressed as the load divided by surface area of indentation.







Vickers Hardness

- ❑ It uses square-base diamond pyramid indenter.
- ❑ The included angle between opposite faces of the pyramid is 136° .
- ❑ This angle was chosen because it approximates the most desirable ratio of indentation diameter to ball diameter in the Brinell hardness test.
- ❑ Because of indenter shape, Vickers hardness number also called as diamond pyramid hardness number (DPH)
- ❑ The loads used with this test range from 1 to 120kg.





Rockwell Hardness Test

- ❑ Most widely used hardness test in USA
- ❑ Its general acceptance is due to its
 - speed, freedom from personal error, ability to distinguish small hardness differences in hardened steel, and the small size of the indentation.
- ❑ Therefore finished heat-treated parts can be tested without damage.



Rockwell Hardness Test

- ❑ This test utilizes the depth of indentation, under constant load, as a measure of hardness.
- ❑ A minor load of 10 kg is first applied to seat the specimen.

This minimizes the amount of surface preparation needed and reduces the tendency for ridging or sinking in by the indenter.
- ❑ The major load is then applied, and the depth of indentation is automatically recorded on a dial gage in terms of arbitrary hardness numbers.
- ❑ Brinell and Vickers hardness have units of kg/mm^2 , however the Rockwell hardness numbers are purely arbitrary.



Rockwell Hardness Test

- ❑ As combination of load and indenter will not produce satisfactory results for materials with a wide range of hardness, following indenters are used in Rockwell Hardness Test

A 120° diamond cone with a slightly rounded point, called a Brale indenter, and 1.6 and 3.2 mm diameter steel balls

- ❑ Major loads of 60, 100, and 150 kg are used.
- ❑ Since the Rockwell hardness is dependent on the load and indenter, it is necessary to specify the combination which is used.



Rockwell Hardness Test

- ❑ Therefore Rockwell Hardness number is prefixed with a letter.

The letter indicates the particular combination of load and indenter for the hardness scale employed.

- ❑ A Rockwell hardness number without the letter prefix is meaningless.
- ❑ Hardened steel is tested on the **C scale** with the diamond indenter and a 150-kg major load.

The useful range for this scale is from about RC 20 to RC 70.



Rockwell Hardness Test

- ❑ Softer materials are usually tested on the **B scale** with a 1.6 mm diameter steel ball and a 100-kg major load.

The range of this scale is from RB 0 to RB 100.

- ❑ The **A scale**, (diamond penetrator & 60-kg major load) provides the most extended Rockwell hardness scale, which is usable for materials from annealed brass to cemented carbides.
- ❑ Many other scales are available for special purposes.





Microhardness Tests

- ❑ Many metallurgical problems require the determination of hardness over very small areas.

The measurement of the hardness gradient at a carburized surface, the determination of the hardness of individual constituents of a microstructure, or the checking of the hardness of a delicate watch gear might be typical problems.

- ❑ The development of the Knoop indenter by the National Bureau of Standards and the introduction of the Tukon tester for the controlled application of loads down to 25 g have made microhardness testing a routine laboratory procedure.



Microhardness Tests

- ❑ The Knoop indenter is a diamond ground to a pyramidal form that produces a diamond-shaped indentation with the long and short diagonals in the approximate ratio of 7 : 1 resulting in a state of plane strain in the deformed region.
- ❑ The Knoop hardness number (KHN) is the applied load divided by the unrecovered projected area of the indentation.

$$KHN = \frac{P}{A_p} = \frac{P}{L^2 C}$$

where P = applied load, kg

A_p - unrecovered projected area of indentation, mm²

L = length of long diagonal, mm

C = a constant for each indenter supplied by manufacturer.



Microhardness Tests

- ❑ The special shape of the Knoop indenter makes it possible to place indentations much closer together than with a square Vickers indentation, e.g., to measure a steep hardness gradient.
- ❑ Its other advantage is that for a given long diagonal length the depth and area of the Knoop indentation are only about 15 percent of what they would be for a Vickers indentation with the same diagonal length.
- ❑ This is particularly useful when measuring the hardness of a thin layer (such as an electroplated layer), or when testing brittle materials where the tendency for fracture is proportional to the volume of stressed material.



Microhardness Tests

- ❑ The low load used with microhardness tests requires that extreme care be taken in all stages of testing.
- ❑ The surface of the specimen must be carefully prepared.
- ❑ Metallographic polishing is usually required.
- ❑ Work hardening of the surface during polishing can influence the results.
- ❑ The long diagonal of the Knoop impression is essentially unaffected by elastic recovery for loads greater than about 300 g.
- ❑ However, for lighter loads the small amount of elastic recovery becomes appreciable.

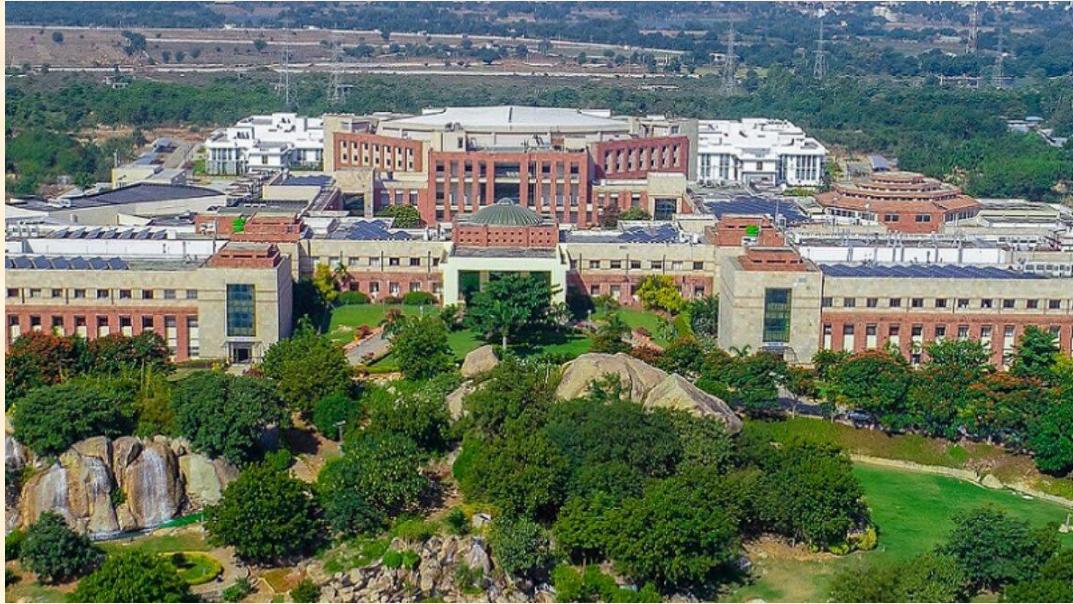


Microhardness Tests

- ❑ Further, with the very small indentations produced at light loads the error in locating the actual ends of the indentation become greater.
- ❑ Both these factors have the effect of giving a high hardness reading.
- ❑ Therefore it is usually observed that the Knoop hardness number increases as the load is decreased below about 300 g.



Significance of Tensile Testing



*LECTURE DELIVERED TO STUDENTS OF
BITS, PILANI (Sep 2026)*

Dr. V.P. Raghupathy

Professor (Retired), Dep. of Mech. Engg., PES University, Bangalore

Classification of Materials

Classification based on arrangement of atoms

- ❑ Metals and Alloys
- ❑ Ceramics, Glasses and Glass-ceramics
- ❑ Polymers (plastics), Thermoplastics and Thermosets
- ❑ Semi-conductors
- ❑ Composite Materials

Classification based on Service Function

- ❑ Structural Materials
- ❑ Aerospace Materials
- ❑ Biomedical Materials
- ❑ Electronic Materials
- ❑ Energy Technology and Environmental Technology
- ❑ Magnetic Materials
- ❑ Photonic or Optical Materials
- ❑ Smart Materials

Example of Materials

❑ Metals and Alloys

E.g. Gray cast iron

Application : Automobile engine blocks

Properties : Easily Castable & machinable, vibration damping

❑ Ceramics, Glasses, and Glass-ceramics

E.g. SiO_2 - Na_2O - CaO :

Application : Window glass

Properties : Optically transparent, thermally insulating

❑ Polymers (plastics), Thermoplastics and Thermosets

E.g. Polyethylene

Application : Food packaging

Properties : Easily formed into thin, flexible, airtight film

❑ Semiconductors

E.g. Silicon

Application : Transistors and integrated circuits

Properties : Unique electrical behavior

❑ Composite Materials

Classification of Composite Materials

Based on geometrical nature
of reinforcement phase

Composites

Particle
reinforced

Large
Particle

Dispersion
Strengthened

Fiber
reinforced

Continuous
(aligned)

Discontinuous
(Short)

Aligned

Random

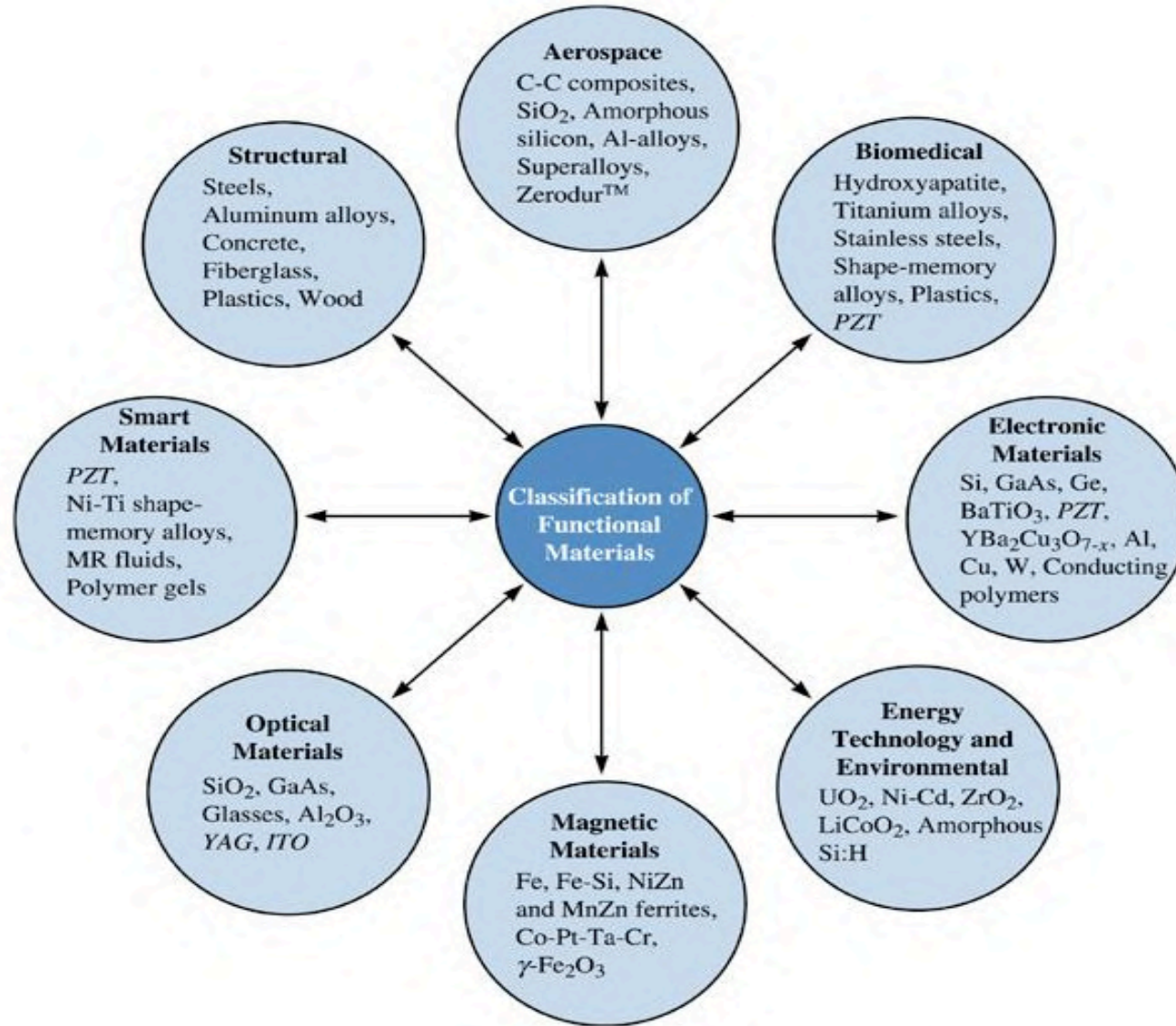
Structural

Laminates

Sandwich
Panels

Classification of Materials based on service functionality

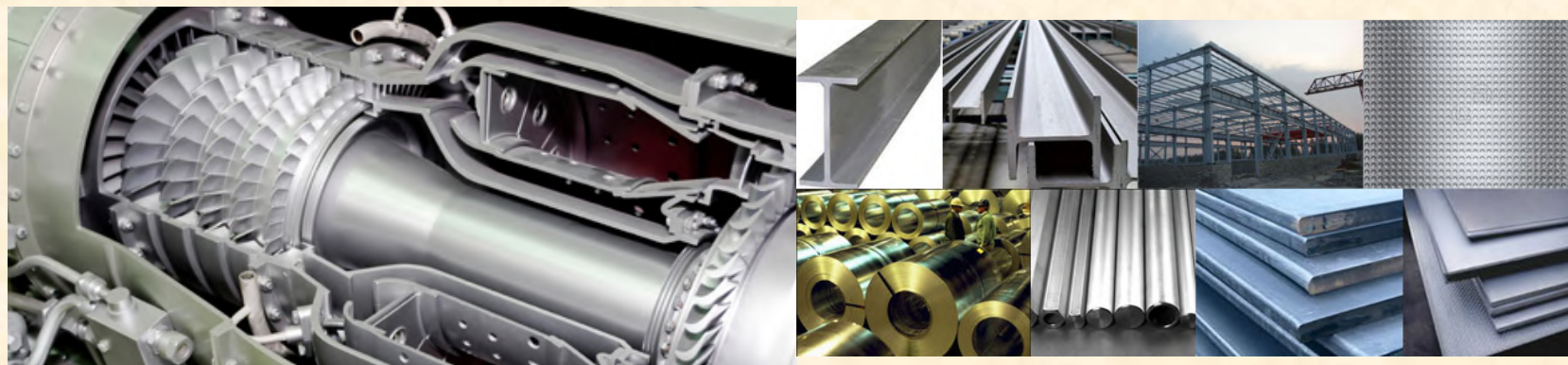
Classification – Based on Functions



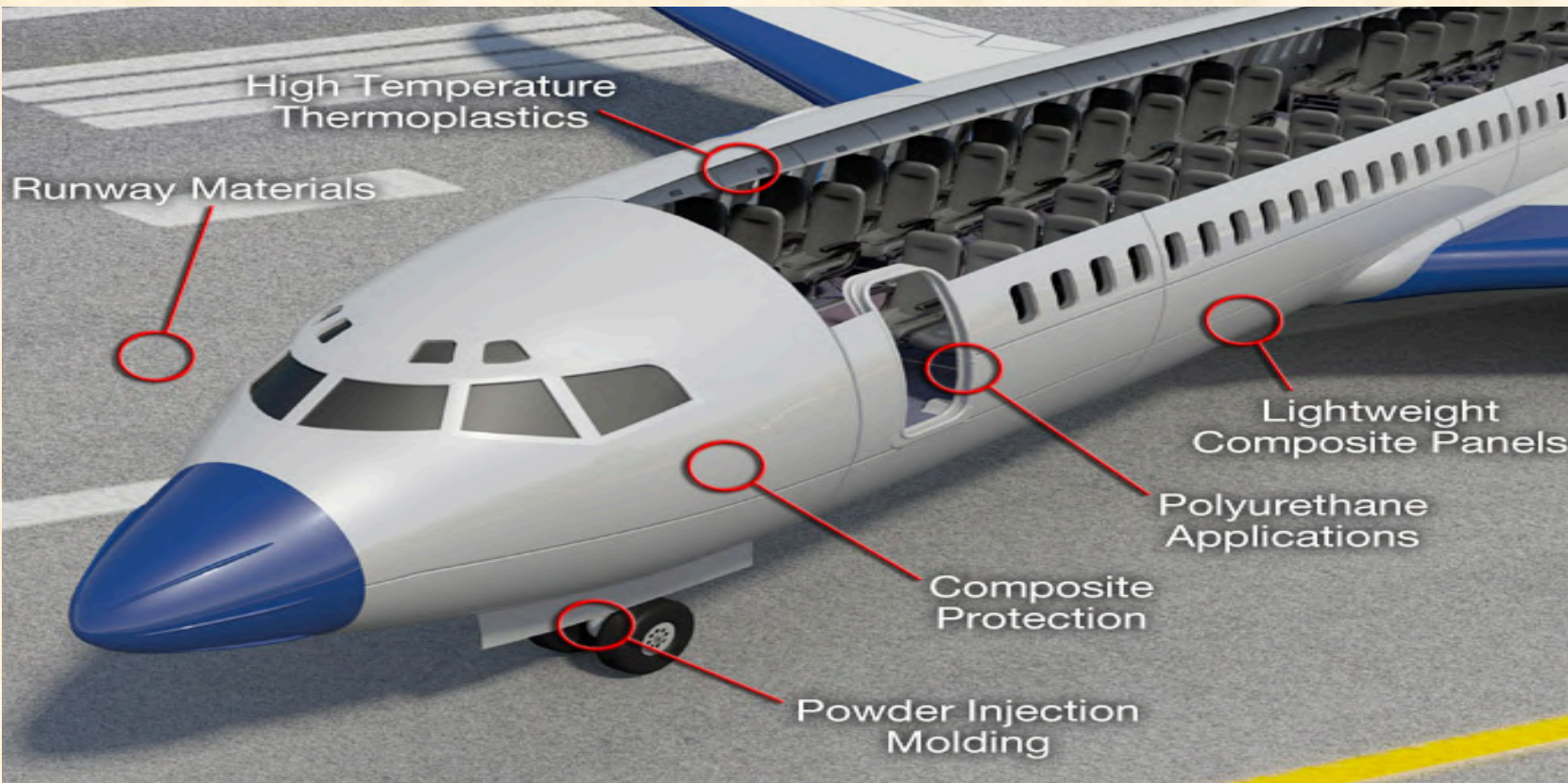
Structural Materials....

Structural materials are materials used or studied primarily for their **mechanical properties**, as opposed to their electronic, magnetic, chemical or optical characteristics. This can include a materials response to an applied force, whether this response is elastic or plastic, its hardness, and its strength.

- Wrought iron: Cast iron
- Steel: Stainless steel.
- Concrete: Reinforced concrete
- Pre-stressed concrete.
- Aluminum : Plastics
- Composites :Timber



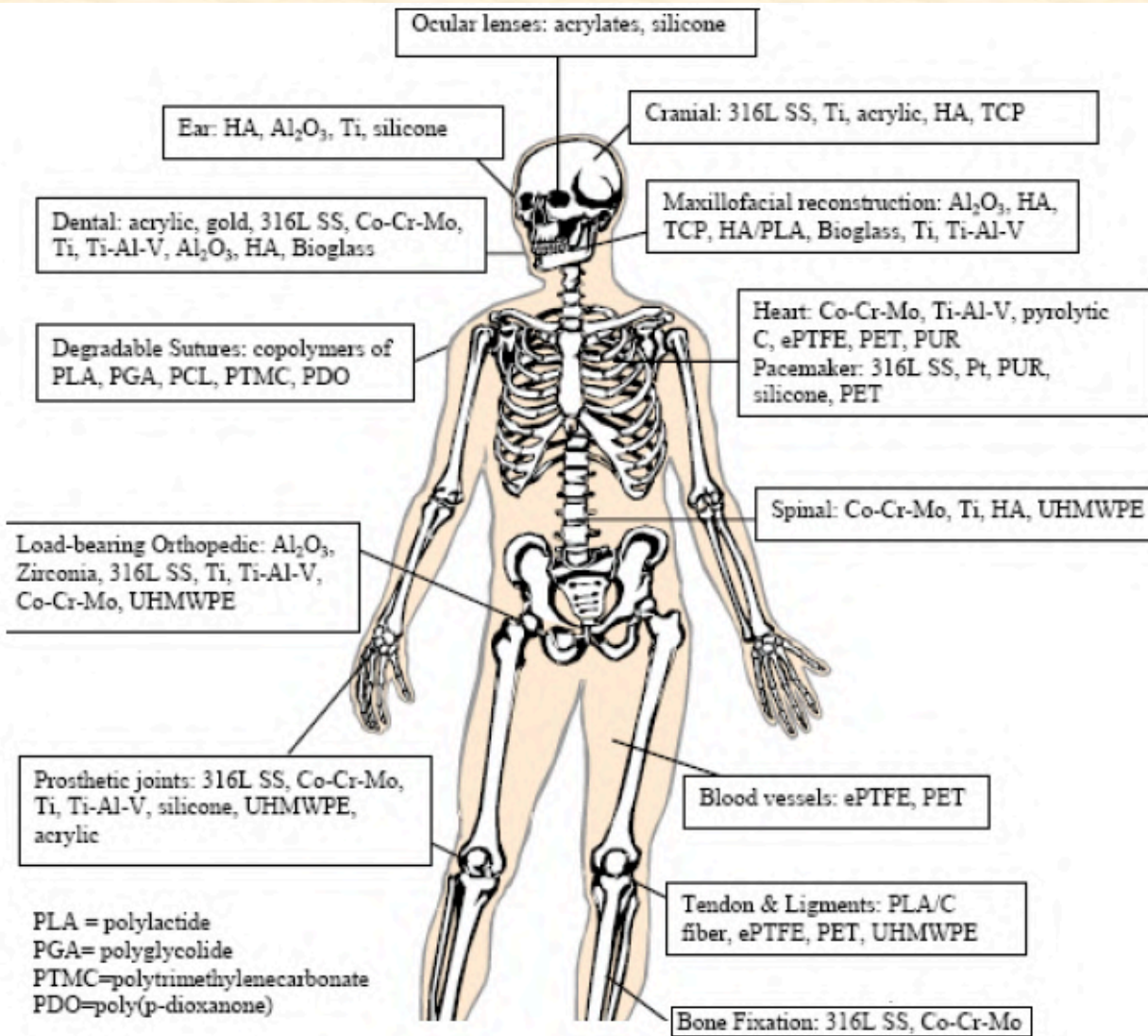
Aerospace Materials....



Aerospace materials - light-weight designs without compromising integrity

Characteristics required : high modulus; high strength to weight ratio ; impact resistance ; fire-resistant and recyclable, dimensional stability, noise and vibration resistance ; high strength concrete systems for runways.

Bio-Medical Materials....

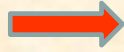


Compatibility
with blood
vessels, bone
minerals and
hormones etc



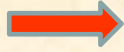
Hip Joints
Orthopaedic
implants
Titanium alloy
Co – Cr – Mo
alloy

Nuclear Sector



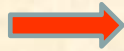
UO₂ , Glasses, Stainless Steel,
Plutonium etc
Managing Nuclear waste

Automotive Sector



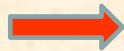
Fuel cells & Batteries - ZrO₂
Managing Industrial waste

Computer Applications



Frames & Batteries – Ceramics,
Polymers
Managing soil Pollution

Petroleum Sector

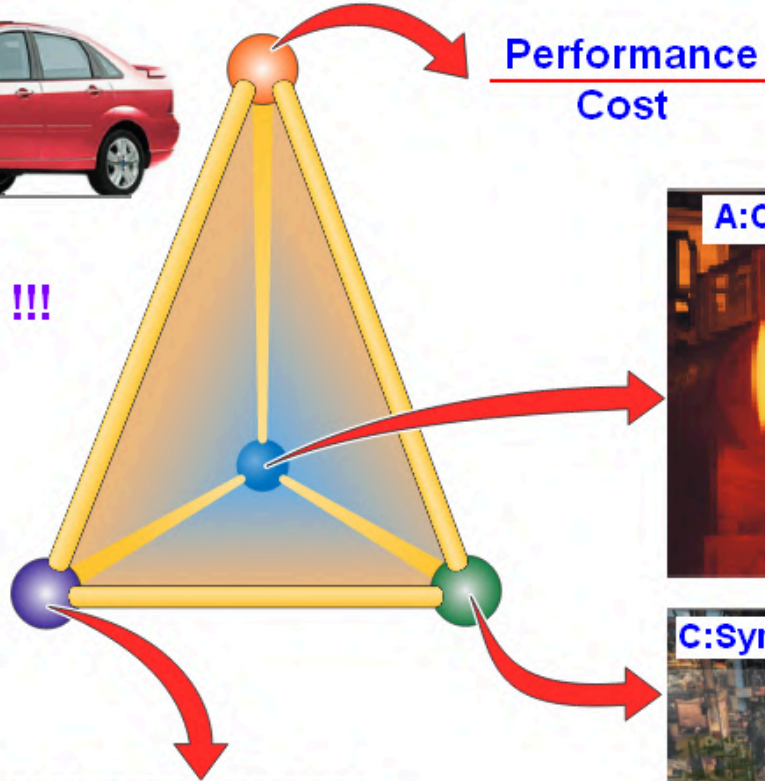


Catalysts in refineries – Zeolites,
Alumina substrates: Pt/Rh as
catalysts
Managing Toxic substances



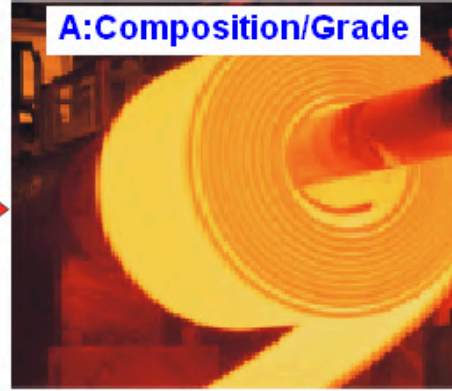
Engineers concern !!!

- Composition
- Weight
- Strength
- Toughness
- Malleability
- Durability



Performance
Cost

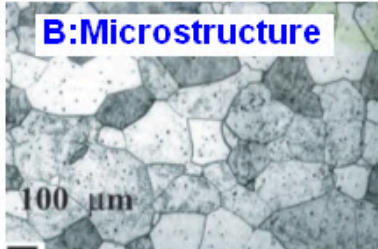
A: Composition/Grade



C: Synthesis/Processing



B: Microstructure



Concept of stress and strain

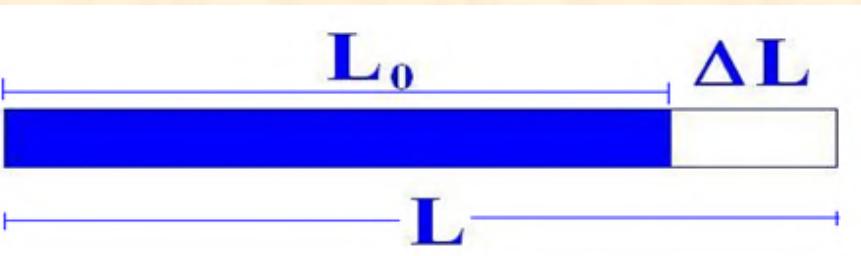
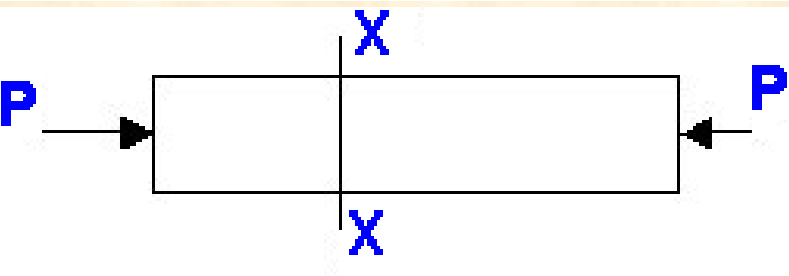
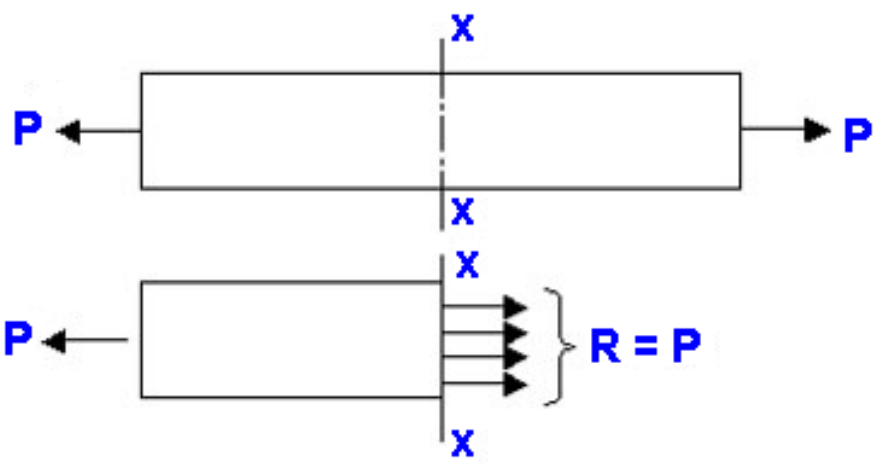
Concept of stress & strain

Tensile stress ($\sigma = P / A$)

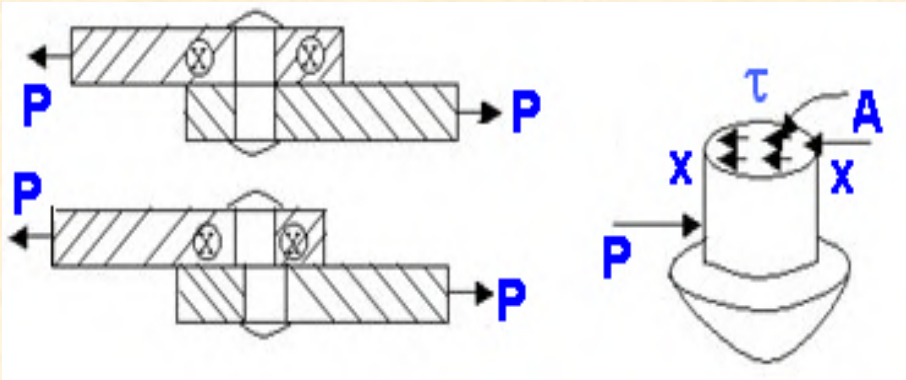
Compressive stress ($\sigma = P / A$)

Normal strain

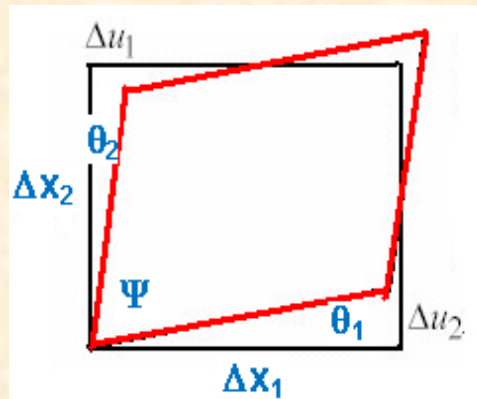
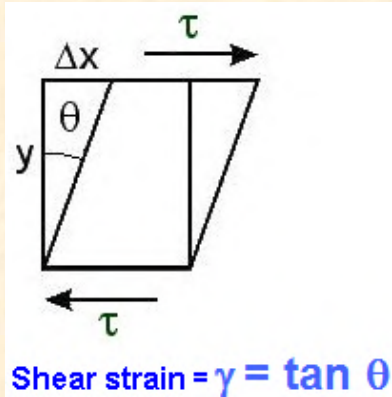
$$\epsilon = \frac{L - L_0}{L_0} = \frac{\Delta L}{L_0}$$



Concept of stress & strain



$$\text{Shear stress } (\tau = P / A)$$



$$\text{Shear strain} = (\gamma / 2)$$

$$\gamma / 2 = \tan \theta_1 + \tan \theta_2$$

Normal strains cause changes in dimensions and shear strains cause changes in shape

Concept of stress & strain

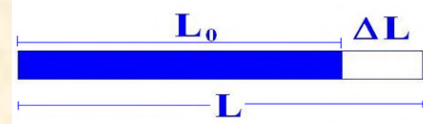
Normal stress causes normal strain and shear stress causes shear strain. This is true only in the case of isotropic materials (all metals).

Normal stress can cause shear strain and shear stress can cause normal strain in the case of orthotropic materials (composites).

Since shear stress is the one which causes plastic deformation, the focus will now be on “shear stress”.

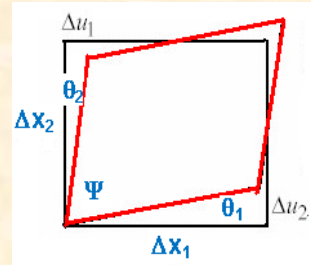
Different modes of strain

Normal strain is produced by compressive or tensile stresses.



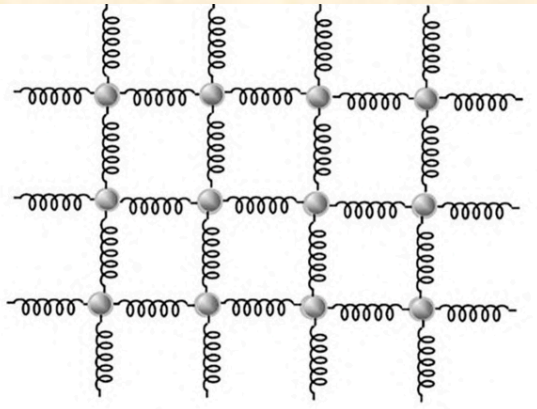
$$\epsilon = \frac{L - L_0}{L_0} = \frac{\Delta L}{L_0}$$

Shear strain is produced by shear stress and occurs as a result of a change in the orientation of the object's molecules.

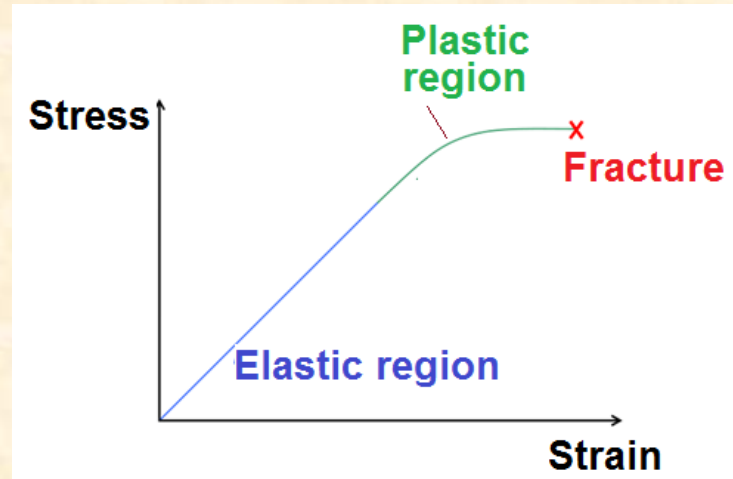


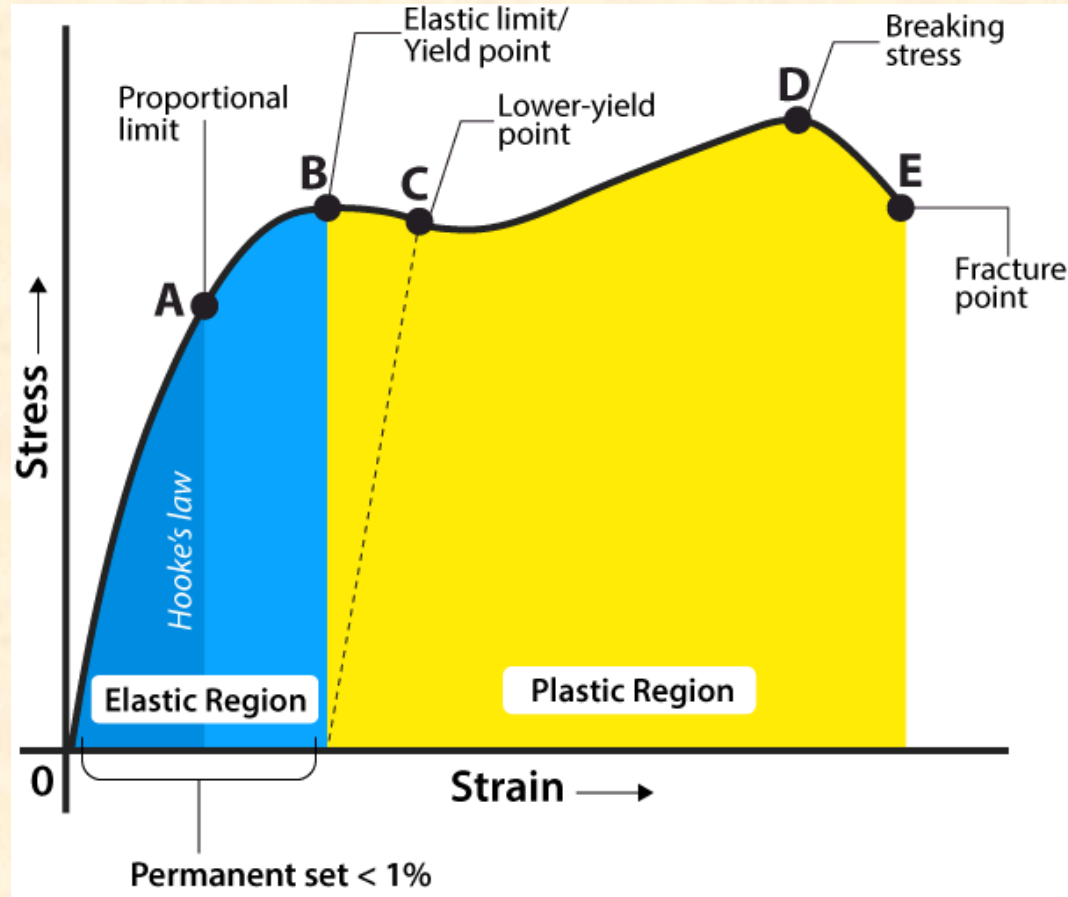
$$\gamma/2 = \tan \theta_1 + \tan \theta_2$$

Elastic strain is **small, time independent and temporary**. It is fully recoverable when the applied load is removed.



Springs represent inter-atomic force



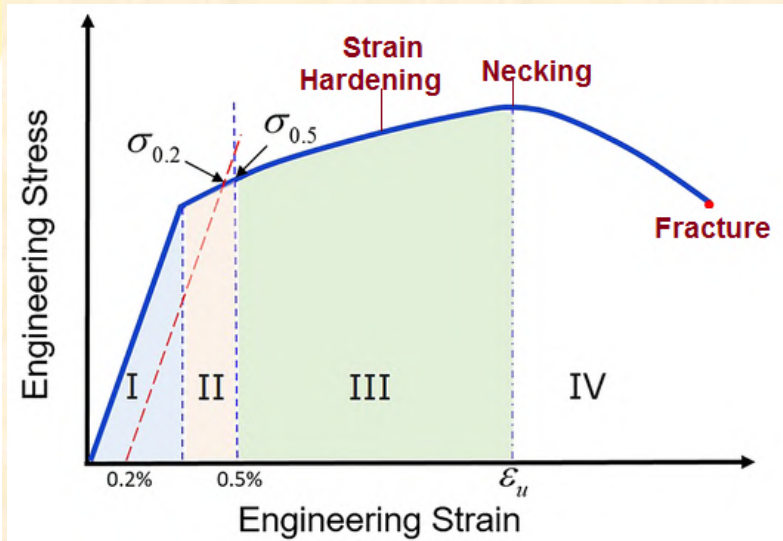
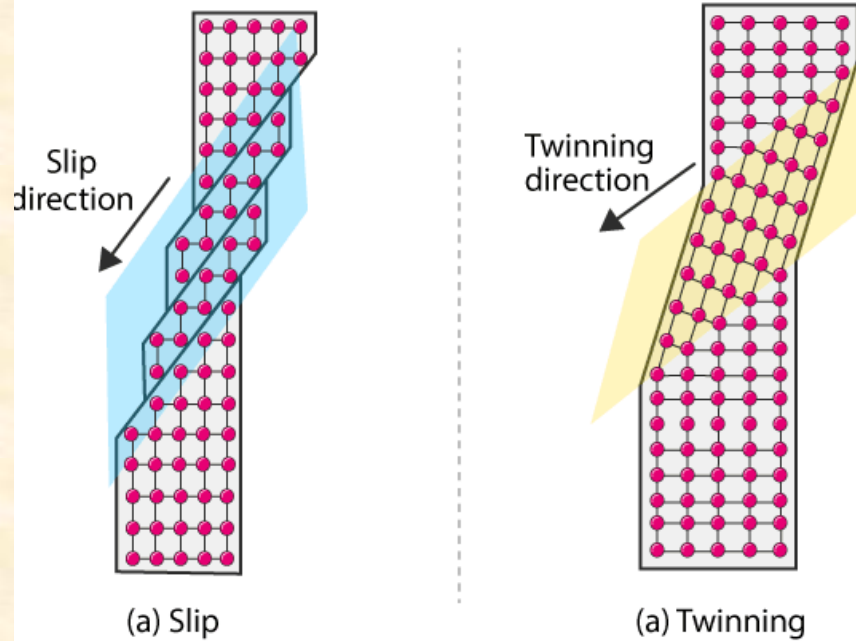


Elastic deformation – small,
Temporary & time independent

Plastic deformation – Large,
Permanent & time
independent

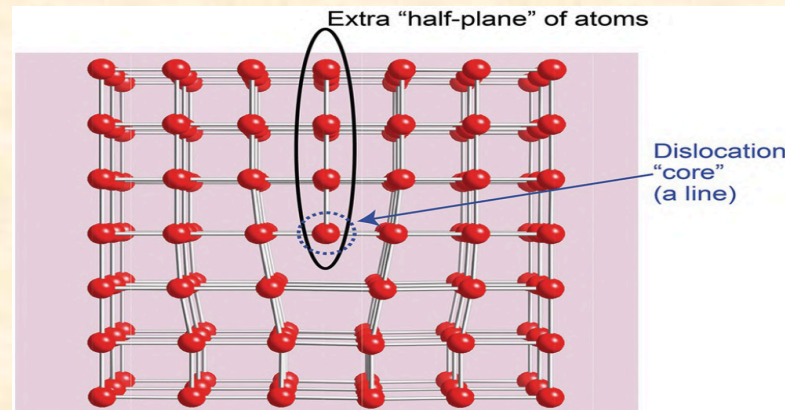
Metallurgy related to Metal Forming

Plastic deformation



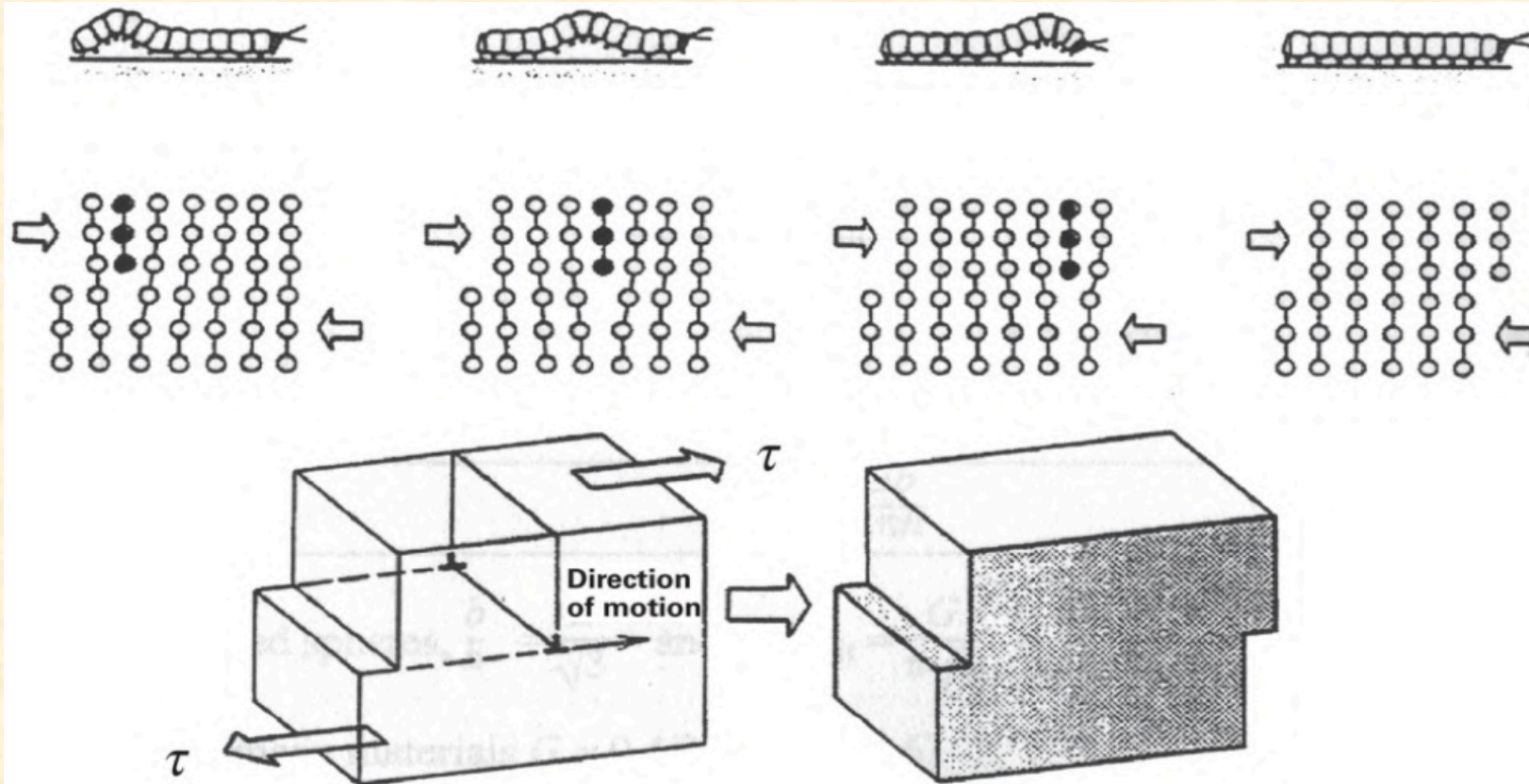
Both Slip & Twinning caused by Shear stress

Plastic strain is **large, time independent and permanent**. It is not recoverable when the applied load is removed.



Movement of dislocations

Dislocations create plastic deformation by moving along their glide planes in response to externally applied stresses. This motion is (1) easy compared with the shift of entire lattice plane at once, and (2) conserves material volume.



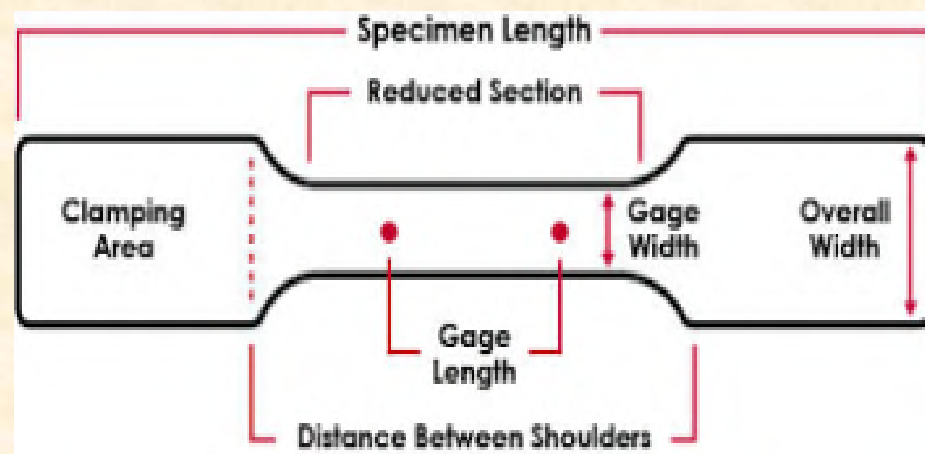
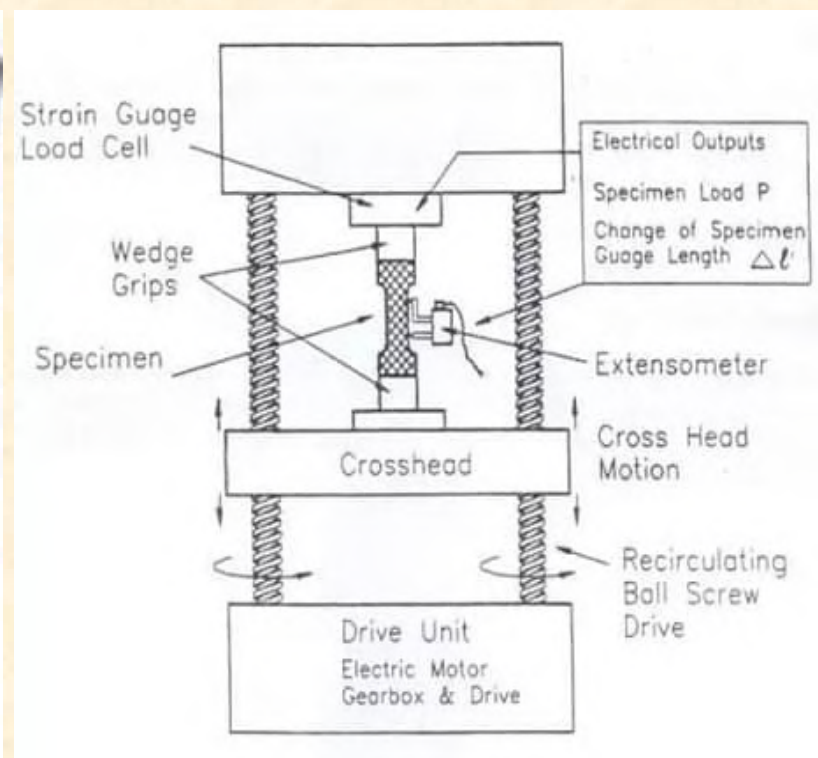
Slip	Twin
Orientation across the slip plane is same	Orientation across the twin plane is different
Atomic movements are equal to atomic distances	Atomic movements are lesser than atomic distances
Atoms are moving in only one plane (slip plane)	Atoms are moving in all planes in the region of twin
Takes place in milli seconds	Takes place in less than micro seconds
Takes place at low strain rates	Takes place at high strain rates
No sound is created	A click sound (Tin cry)

Mechanical Properties

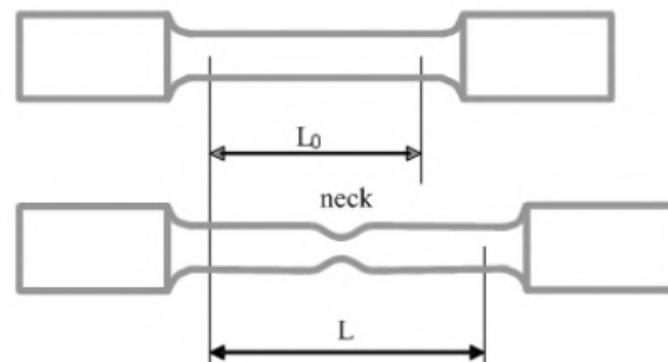
- **Strength**
- **Elasticity**
- **Plasticity**
- **Ductility**
- **Malleability**
- **Brittleness**
- **Stiffness**

- **Hardness**
- **Creep**
- **Fatigue**
- **Resilience**
- **Toughness**

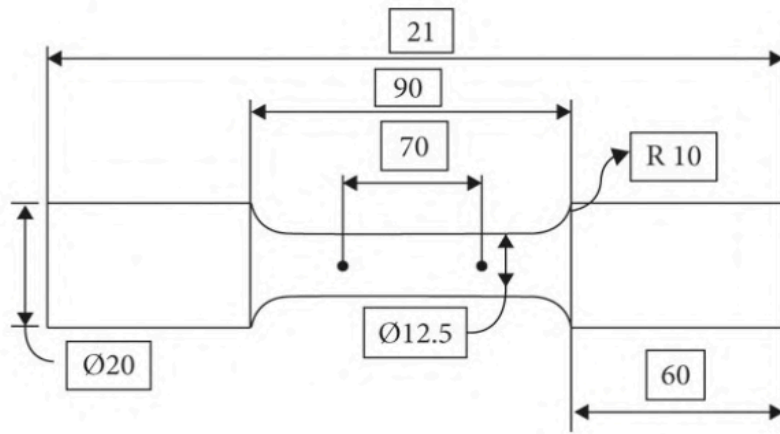
Tensile Testing



Standard tensile test specimen



ASTM Standard E8

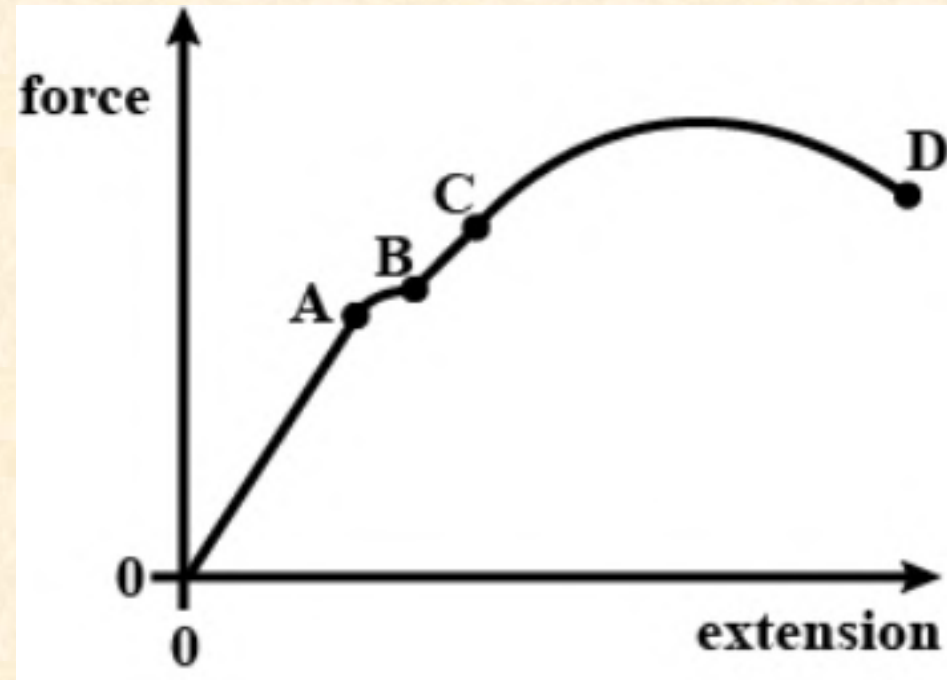


Force or Load obtained from Load cell

Extension is obtained from Extensometer



Extensometer attached to a steel specimen

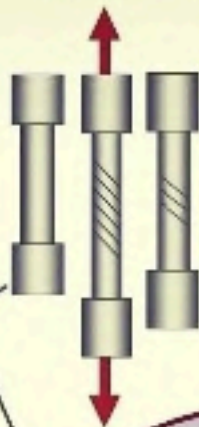


Stress – Strain Curve

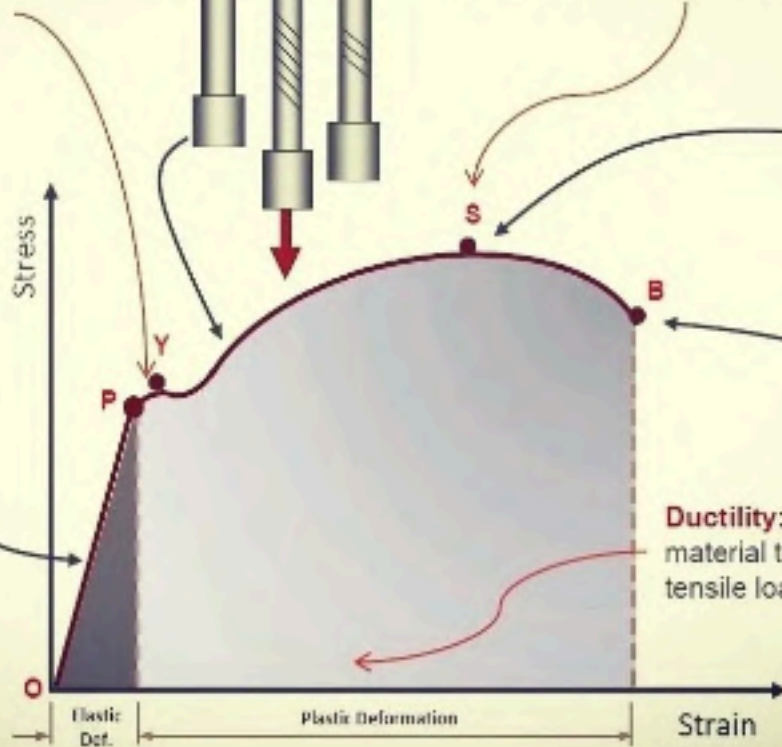
Yield Strength (YS): The stress at which the plastic deformation becomes noticeable

Ultimate Tensile Strength (UTS): The stress that corresponds to the maximum load in the tensile test

Ductility: The ability of a material to deform under tensile load without rupture



P: Elastic Limit
Y: Yield Point
S: Max. Load Value
B: Breaking Point



Tensile Properties

Yield strength

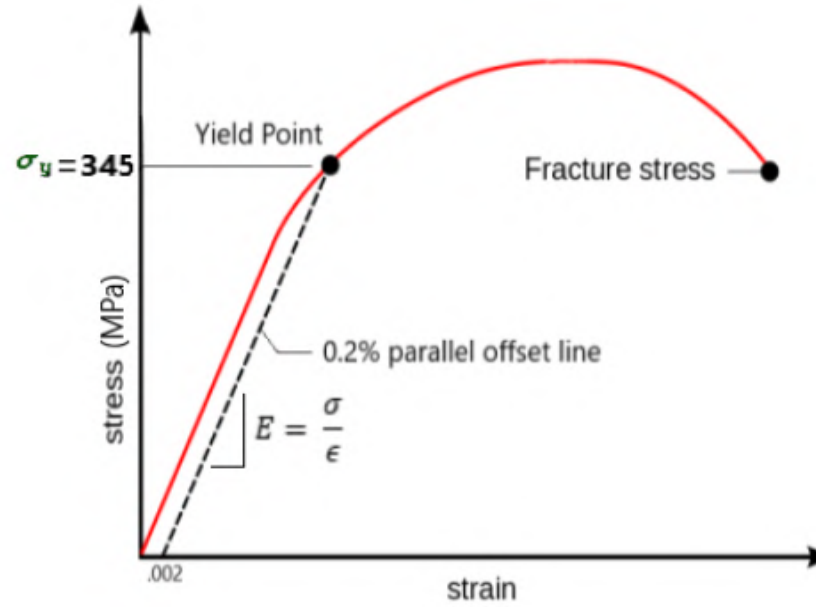
Ultimate Tensile strength

Ductility

Resilience

Strain Hardening Exponent

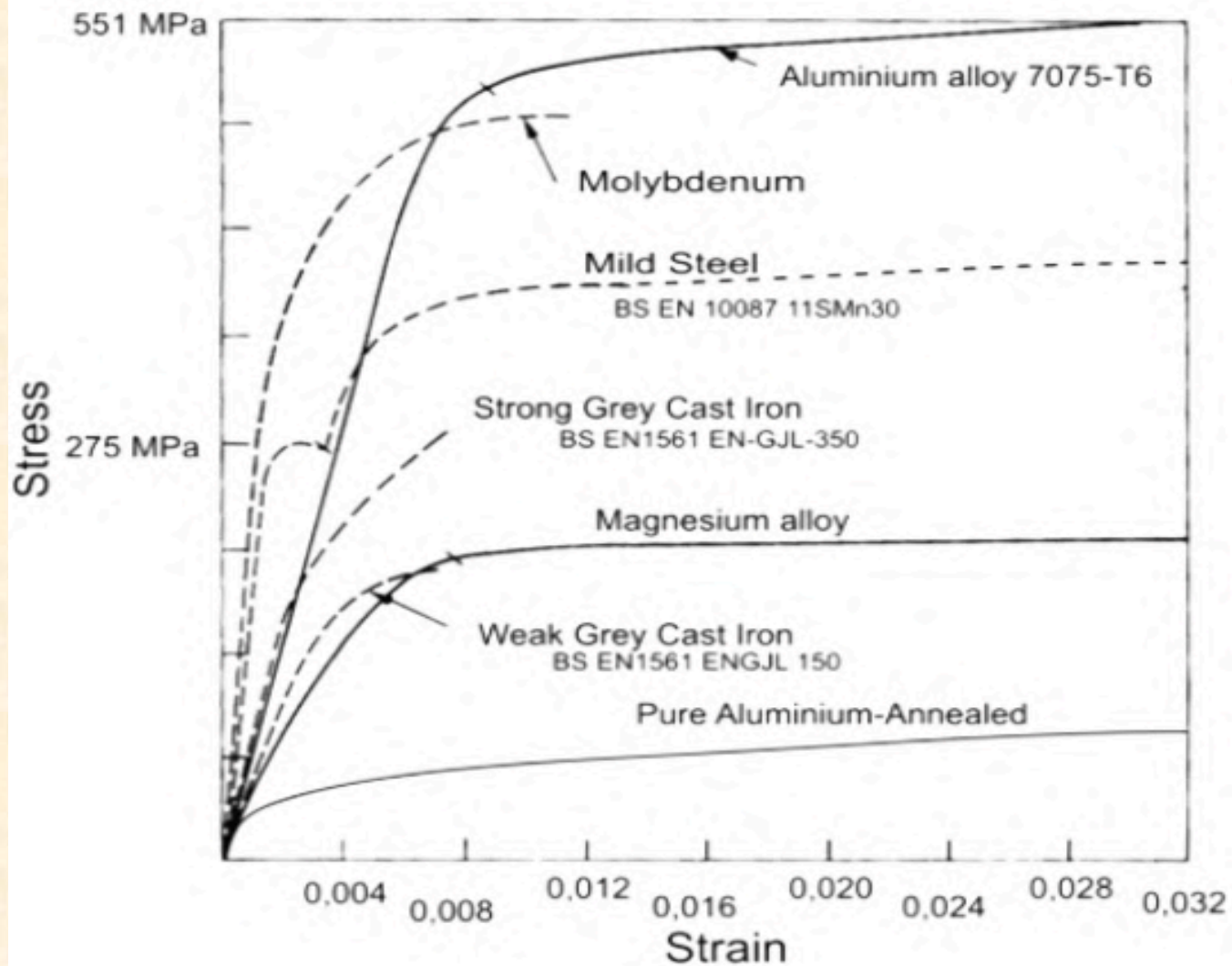
Strain rate sensitivity exponent

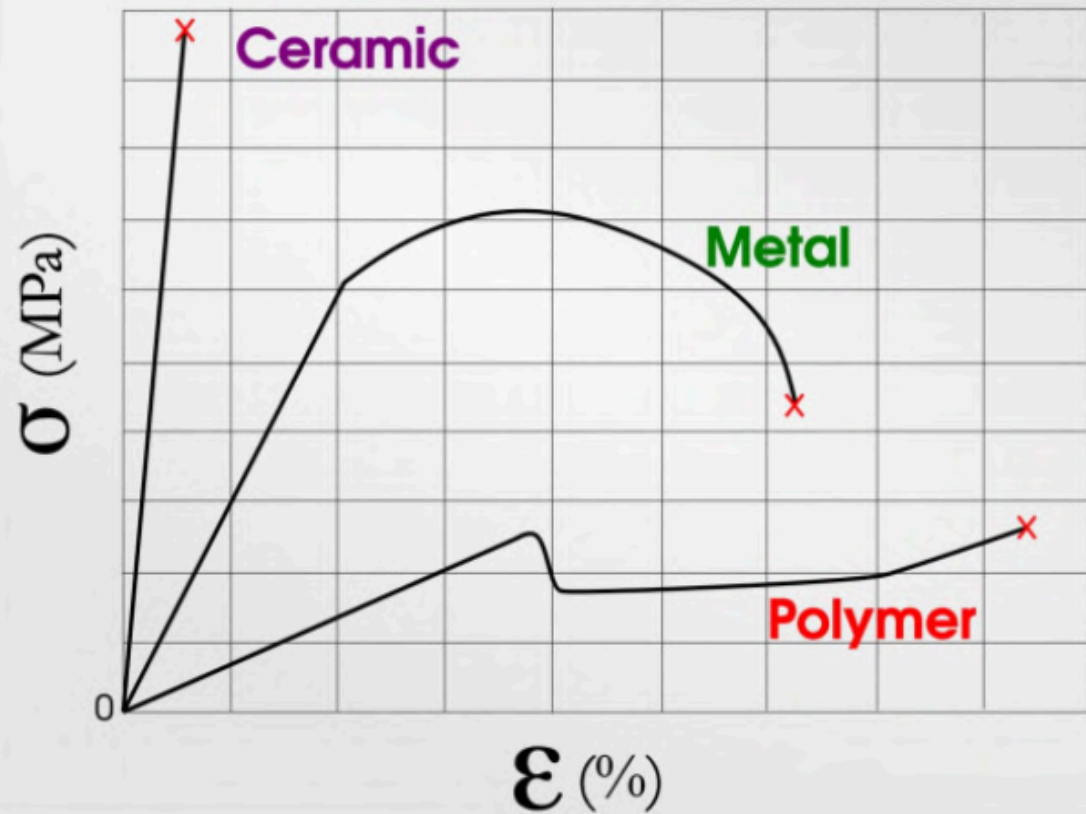


$$\text{Yield strength} = \frac{\text{Yield Force}}{\text{Original cross sectional Area}}$$

$$\text{Ultimate Tensile Strength} = \frac{\text{Max. Force}}{\text{Original cross sectional Area}}$$

$$\text{Ductility} = \frac{\text{Final Length} - \text{Original length}}{\text{Original length}} \times 100$$



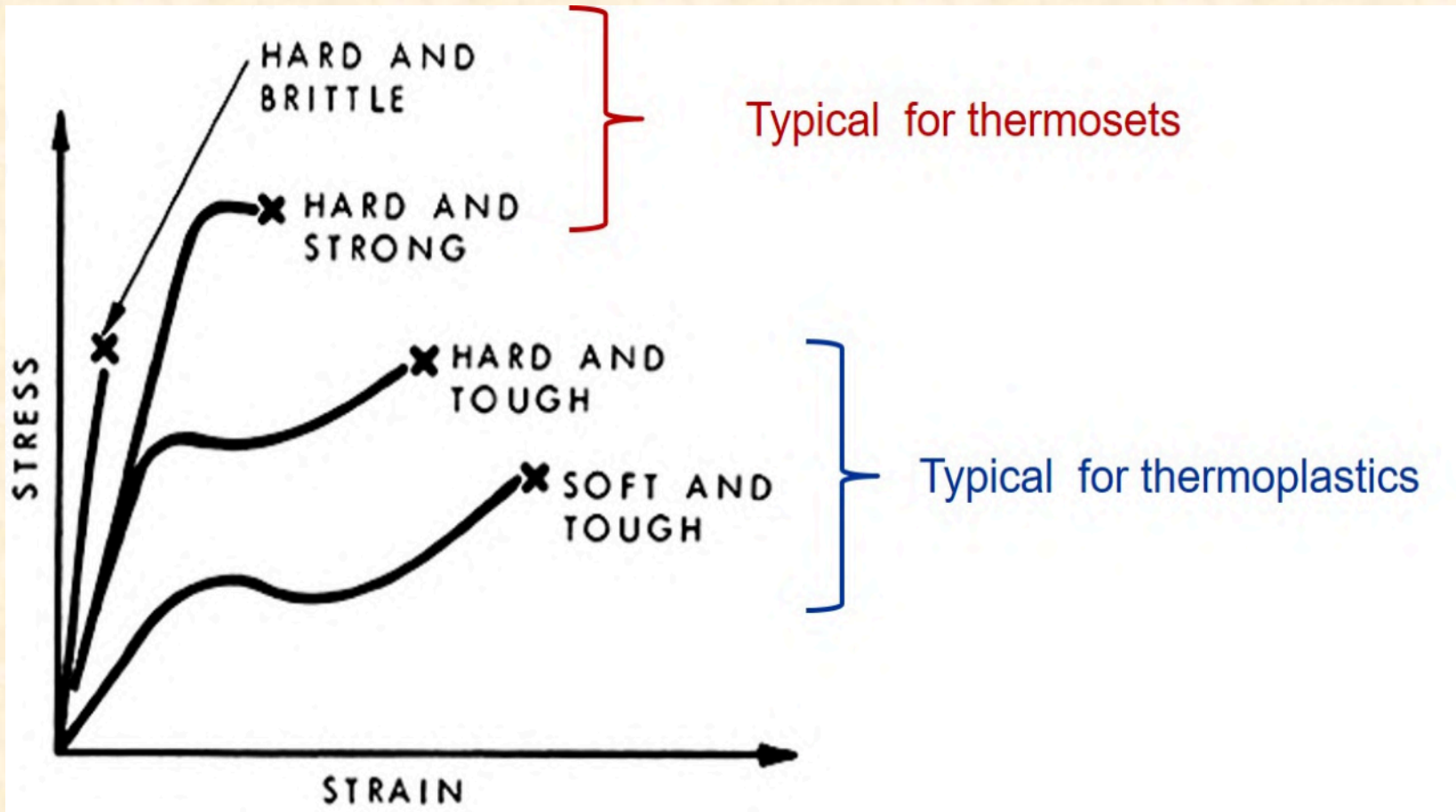


Stress-Strain
Curve for:

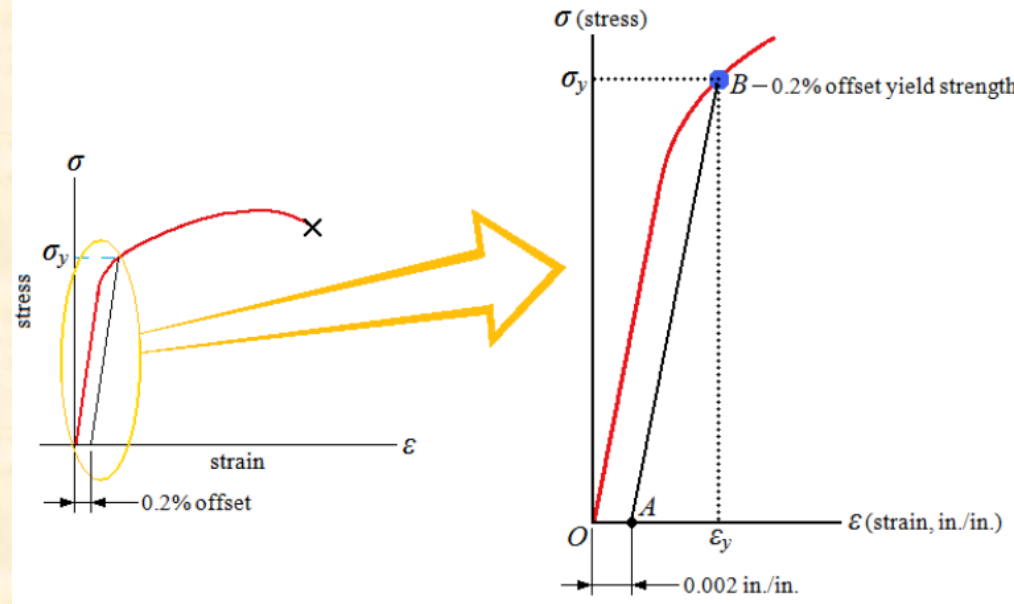
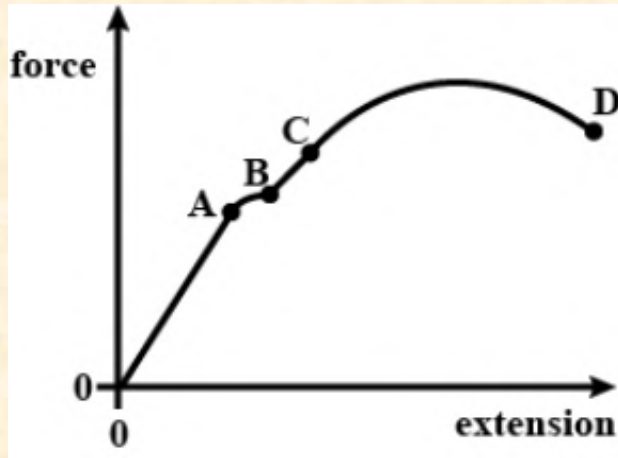
Ceramic

Metal

Polymer



Computations from Load-Extension curve



$$\text{Engg. Stress } (S) = \frac{\text{Load}}{A_0}$$

L_0 : Original length ;

A_0 : Original cross sectional area ;

$$\text{Engg. Strain } (e) = \frac{L - L_0}{L_0}$$

L : Instantaneous length

A : Instantaneous cross sectional area

$$\text{True Stress } (\sigma) = \frac{\text{Load}}{A} = S(1 + e)$$

$$\text{True strain } (\epsilon) = \log_e(1 + e)$$

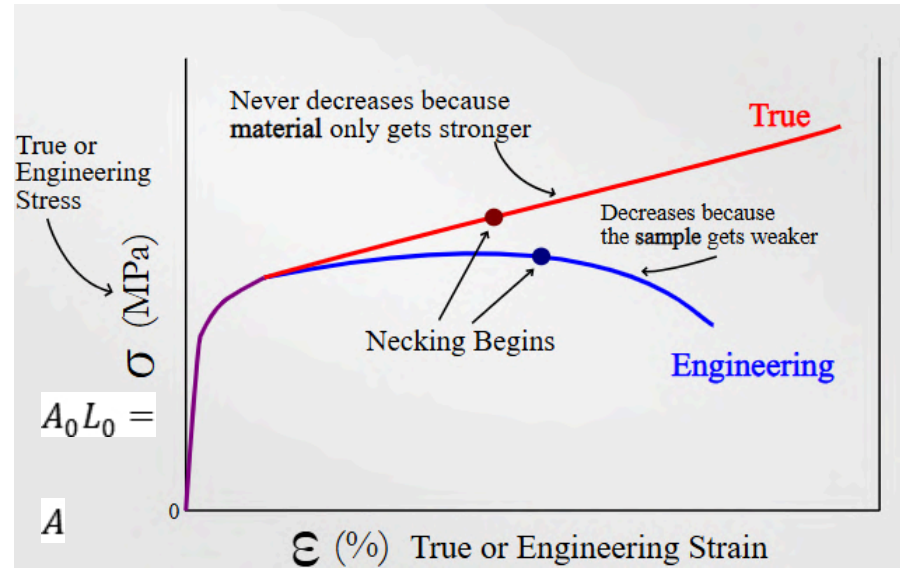
True stress - True strain

$$\sigma_{eng} = \frac{F}{A_0} \quad \varepsilon_{eng} = \frac{\Delta L}{L_0} = \frac{L_f - L_0}{L_0}$$

$$\sigma_{True} = \frac{F}{A_i} \quad \varepsilon_{True} = \ln \left(\frac{L_i}{L_0} \right)$$

True stress is obtained by dividing applied load by instantaneous area

True strain is obtained by integrating incremental engineering strain bet. the limits of initial length to final length



For true stress:

$$\sigma_t = \frac{P}{A} = \frac{P}{A_0} \frac{A_0}{A} = \frac{P}{A_0} \frac{L}{L_0} = \sigma(1 + e)$$

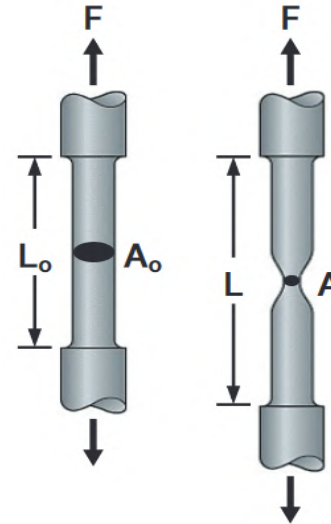
For true strain:

$$\delta \varepsilon_t = \frac{\delta L}{L}$$

$$A_0 L_0 = A L \text{ (Since volume is constant)}$$

Integrate both sides and apply the boundary condition,

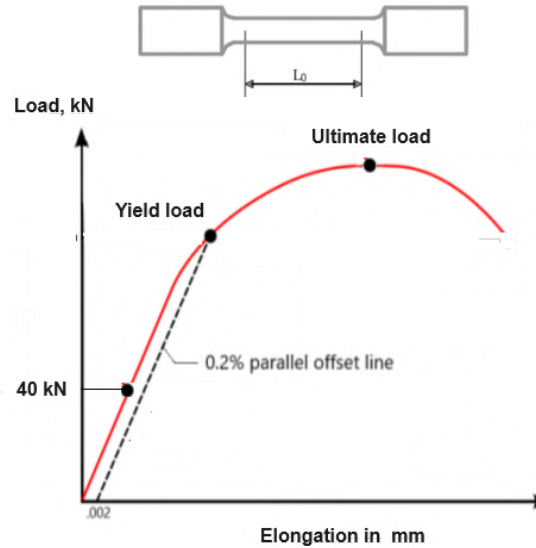
$$\varepsilon_t = \ln \left(\frac{L}{L_0} \right) = \ln(1 + e)$$



Problems in Tensile Testing

Tensile test was performed and following are the data

- Original diameter (d_0): 40 mm
- Original gauge length (L_0): 200 mm
- Elongation at a load of 40 kN (ΔL): 0.0304 mm (within the elastic limit)
- Yield load (F_y): 161 kN
- Maximum/Ultimate load (F_{max}): 242 kN
- Length at fracture (L_f): 249 mm



Find:

1. Engineering Stress at 40 kN
2. Young's Modulus (E)
3. Yield Point Stress
4. Ultimate Tensile Stress (UTS)
5. Percentage Elongation

Formulas Used

- Original Cross-Sectional Area (A_0): $\frac{\pi}{4} \times d_0^2$
- Engineering Stress (σ): $\frac{F}{A_0}$
- Engineering Strain (e): $\frac{\Delta L}{L_0}$
- Young's Modulus (E): $\frac{\text{Stress}}{\text{Strain}}$ (in the elastic region)
- Percentage Elongation: $\frac{L_f - L_0}{L_0} \times 100$

$$A_0 = \frac{\pi}{4} \times (40 \text{ mm})^2 = 1256.64 \text{ mm}^2$$

Young's Modulus (E)

Stress and strain at 40kN Load

$$\sigma_1 = \frac{40,000 \text{ N}}{1256.64 \text{ mm}^2} = 31.83 \text{ N/mm}^2 \text{ (MPa)}$$

$$e_1 = \frac{0.0304 \text{ mm}}{200 \text{ mm}} = 0.000152$$

$$E = \frac{\sigma_1}{e_1} = \frac{31.83 \text{ N/mm}^2}{0.000152} \approx 209.4 \text{ GPa}$$

Yield Point Stress (σ_y)

$$\sigma_y = \frac{161,000 \text{ N}}{1256.64 \text{ mm}^2} = 128.12 \text{ N/mm}^2 \text{ (MPa)}$$

Ultimate Tensile Stress (σ_u)

$$\sigma_u = \frac{242,000 \text{ N}}{1256.64 \text{ mm}^2} = 192.58 \text{ N/mm}^2 \text{ (MPa)}$$

Percentage Elongation

$$\% \text{ Elong} = \frac{249 - 200}{200} \times 100$$

Problems in Tensile Testing

A mild steel tie member has a cross-sectional area of 6 cm^2 and an initial length of 15 cm . It is subjected to a direct axial tensile load of 318 kN . Laboratory data or the material's engineering stress-strain curve indicates that this specific grade of mild steel begins distinct yielding at a strain of 0.2% (0.002), with a yield stress threshold typically around $250\text{--}300 \text{ MPa}$.

(i) Calculate the induced engineering stress (σ) in the member.

(ii) Determine whether the material has entered the plastic/yielding zone or remains elastic.

Step-by-Step Solution

Cross-sectional area (A): $A = 6 \text{ cm}^2 = 6 \times 10^{-4} \text{ m}^2 = 600 \text{ mm}^2$

Tensile force (P): $P = 318 \text{ kN} = 318,000 \text{ N}$ $\epsilon = 0.2\% = 0.002$

Calculate the Applied Engineering Stress (σ)

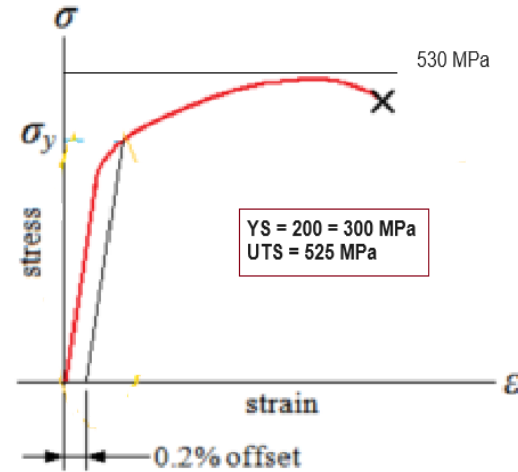
Engineering stress : applied load divided by the **original** cross-sectional area (A_0)

$$\sigma = \frac{P}{A_0} = \frac{318,000 \text{ N}}{600 \text{ mm}^2} = 530 \text{ MPa}$$

Analyze Behavior via the Stress-Strain Curve

Compare the calculated operational stress (530 MPa) against standard mild steel yield stress limits ($\sigma_y \approx 250\text{--}300 \text{ MPa}$) and ultimate tensile strength of 525 MPa .

Because 530 MPa exceeds the ultimate strength and standard yield bounds of regular structural mild steel, the computed stress indicates that an actual member of this size would either experience extreme plastic deformation, necking, or structural failure/rupture



A metal specimen has an initial diameter of 12.7 mm. During a tensile test, a load of 76 kN is applied, reducing the diameter to 12.0 mm. Find the true stress and true strain.

Solution

Convert units to SI:

1. Load (F) = 76 kN = 76×10^3 N
2. Initial diameter (d_0) = 12.7 mm = 12.7×10^{-3} m
3. Final diameter (d) = 12.0 mm = 12.0×10^{-3} m

Calculate initial and final areas (A_0 and A):

$$A_0 = \frac{\pi d_0^2}{4} = \frac{\pi (12.7 \times 10^{-3})^2}{4} \approx 1.27 \times 10^{-4} \text{ m}^2$$

$$A = \frac{\pi d^2}{4} = \frac{\pi (12.0 \times 10^{-3})^2}{4} \approx 1.13 \times 10^{-4} \text{ m}^2$$

Calculate True Stress (σ_t):

$$\sigma_t = \frac{F}{A} = \frac{76 \times 10^3}{1.13 \times 10^{-4}} \approx 672 \text{ MPa}$$

Calculate True Strain (ϵ_t):

$$\epsilon_t = \ln\left(\frac{A_0}{A}\right) = \ln\left(\frac{1.27 \times 10^{-4}}{1.13 \times 10^{-4}}\right) \approx 0.117$$

Tensile testing of Polymers

Plastic Behavior at Different Strain Rates

Plastics (thermoplastics and glassy polymers) feature longer-range chain entanglements and crystallinity, making them stiffer initially.

- **Low/Quasi-Static Rates:** They often exhibit clear ductile yielding, followed by plastic deformation and chain orientation or strain hardening before finally breaking.
- **High/Dynamic Rates:** Yield stress and tensile strength increase, but the material loses its ability to deform as much.
- **Brittle Transition:** High strain rates restrict the molecular chain mobility required for plastic flow. Ductile plastics can turn brittle, showing a sharp drop in elongation at break and lower impact toughness.

Typical Strength Ranges by Polymer Type

General-Purpose & Standard Plastics:

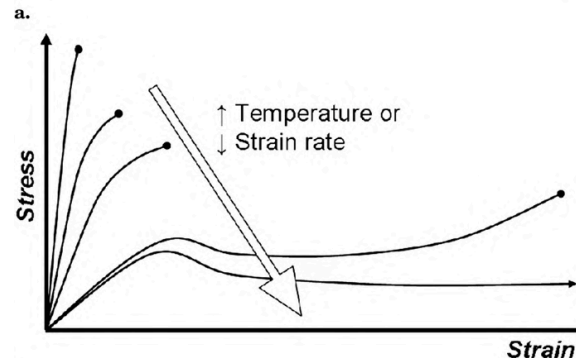
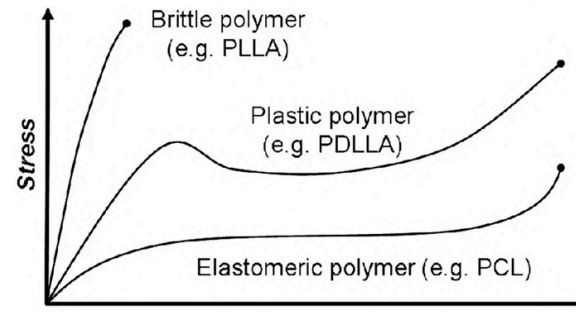
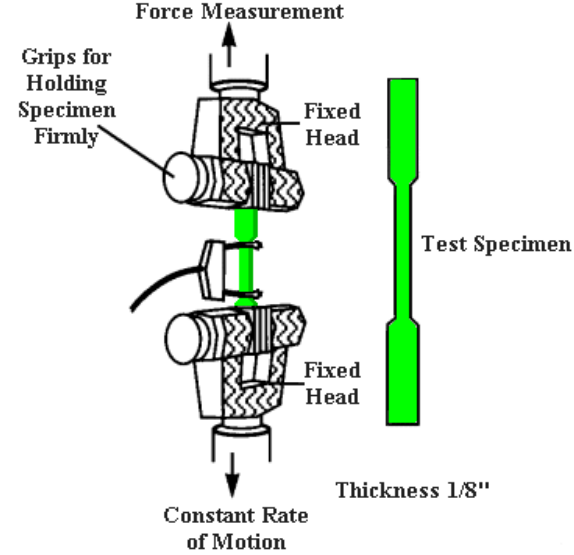
- Polypropylene (PP): **25–35 MPa**
- Polyvinyl Chloride (PVC - rigid): **40–60 MPa**
- Polystyrene (PS): **35–60 MPa**
- High-Density Polyethylene (HDPE): **20–30 MPa**

High-Performance Engineering Polymers:

- Polyetheretherketone (PEEK): **~96 MPa**
- Polyamideimide (PAI): **~145 MPa**

Stiffness (Young's Modulus):

- Typically very low, usually between **0.5 to 4 GPa** for unreinforced plastics, compared to 100+ GPa for metals.



b.